

IOTA TECHNOLOGIES · KINGDOM OF SAUDI ARABIA

Sanad سند

Shariah-Compliant Origination Platform
Wasl وصل — SME Supply Chain Finance

Solution Design Document

Business Requirements · High Level Design · Architecture · Data Design · Backend Design · Frontend Design

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Platform partners	Tuum (core banking) · Nutrient (document & signature)

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Document Control

Version history

Version	Date	Author	Summary of change
0.1	17 Sep 2026	IOTA Solution Architecture	Initial draft. Covers BRD, HLD, Architecture, Data Design, Backend Design and Frontend Design for the Wasl SME Supply Chain Finance product on the Sanad platform.
0.2	17 Sep 2026	IOTA Solution Architecture	Shariah governance confirmed. Both target institutions — an Islamic bank and a fintech — hold sitting Shariah Supervisory Boards. OI-01 closed. Phase 0 shortened and re-scoped from obtaining rulings to harvesting existing ones. New Section 3.11 on multi-board Shariah configuration. New OI-19 on two-board divergence. Revised assumptions, risks, immediate actions and verification register.

Distribution

Role	Purpose
Client executive sponsor	Approval of business scope, phasing and commercial model
Shariah Supervisory Boards (SSB) Both institutions — issued separately	Review and ruling on product structures, contract sequencing, execution pattern and governance model. Sections 2.9, 3.5, 3.6, 3.11 and 5.5 are addressed to the Boards. Each Board rules independently on its own institution's deployment; this document is not a request for a joint ruling.
Head of SME / Corporate Banking	Business requirements, product set, commercial viability
Chief Risk Officer / Credit	Decisioning model, risk controls, limit framework
Chief Technology Officer / Enterprise Architecture	Architecture, integration, deployment, security
Compliance / AML / Data Protection Officer	SAMA CSF, SAMA Cloud Rules, PDPL, AML-CFT
Tuum and Nutrient account teams	Platform capability confirmation and licensing

Status of this document

IMPORTANT — READ BEFORE RELYING ON THIS DOCUMENT

- 1. Shariah authority.** This document is a systems design. It is not a fatwa and carries no Shariah authority. Every product structure, contract sequence, document template and control described here is *subject to the ruling of the deploying institution's Shariah Supervisory Board*. Where this document states a position on Islamic commercial law, it does so to make the engineering requirement explicit — not to settle the jurisprudence. The Board's ruling governs, and where a Board differs from this document, the design changes.
- 1a. Two Boards.** Two institutions are in scope, each with its own sitting Board. They may rule differently on the same question, and the platform is designed to accommodate that rather than to force convergence. Section 3.11 sets out how. No statement in this document should be read as one Board's position being binding on the other.
- 2. Platform capability is unconfirmed.** Specific Tuum and Nutrient API surfaces, and their Islamic-finance and supply-chain-finance capability, have not yet been verified against vendor documentation or a sandbox environment. Statements about vendor capability in this document are *capability-level design intent* and are recorded in the Verification Register (Section 9) as open items. Effort estimates and phase durations are contingent on those answers.
- 3. Regulatory parameters are placeholders.** SAMA thresholds, licensing requirements, ZATCA technical specifications and PDPL obligations must be confirmed against the current published rules with the client's compliance function before build. All such values are held as effective-dated configuration, never as code.

How to read this document

Section 2 — BRD	What the business needs and why. Numbered, testable requirements. Read this if you own the P&L, the credit policy or the regulatory relationship.
Section 3 — High Level Design	How the solution works end to end, the journeys, and the Shariah sequencing model. Read this if you are the SSB, the business, or the architect.
Section 4 — Architecture	Components, integration, security, deployment, resilience. Read this if you are enterprise architecture, security or infrastructure.
Section 5 — Data Design	Domain model, entities, state machines, classification, retention, residency. Read this if you are data, DPO or audit.
Section 6 — Backend Design	Services, APIs, engines, patterns, testing. Read this if you are building it.
Section 7 — Frontend Design	Surfaces, journeys, design system, Arabic-first and RTL, accessibility. Read this if you are building or approving the user experience.
Sections 8–10	Delivery plan, open verification register, glossary.

Requirements are identified by prefix so they can be traced through build and test: **BR** business, **SH** Shariah, **RC** regulatory and compliance, **NFR** non-functional, **INT** integration, **DD** data, **FE** frontend, **BE** backend, **OI** open item.

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1. Executive Summary

Sanad is a Shariah-compliant origination platform. Wasl is its first product: supply chain finance for Saudi small and medium enterprises, built so that a non-compliant transaction is not merely discouraged but structurally impossible to produce.

1.1 The opportunity

Saudi Arabia has a large, well-documented SME credit gap, and closing it is an explicit national objective under Vision 2030. Supply chain finance is the most efficient instrument for closing it, because it underwrites the strength of a large anchor buyer rather than the weakness of a small supplier.

Two conditions make the Kingdom unusually well suited to this product right now:

- **Mandatory e-invoicing.** Under ZATCA's FATOORA regime, business-to-business tax invoices are generated in structured form, cryptographically stamped, and cleared with the tax authority. The three failure modes that historically destroy supply chain finance programmes — fabricated invoices, the same invoice financed at multiple lenders, and unverifiable SME revenue — are substantially solved by national infrastructure that already exists.
- **A mature Islamic finance market.** Supply chain finance is trade finance, and trade is the native domain of Islamic commercial law. Every structure this product needs is unanimously accepted. Compliance is not a tax on this product; it is an advantage.

1.2 The core design idea

Conventional supply chain finance is, in every variant, the purchase of a receivable at a discount. Under the AAOIFI and wider Gulf position this is *bay' al-dayn* — sale of debt — and it is not permitted. Conventional supply chain finance therefore cannot be adapted; it must be replaced.

Wasl replaces it with a real trade. The financier does not buy paper. It buys goods, takes title, bears ownership risk for a genuine and evidenced interval, and sells them on at a disclosed markup under a Murabaha. The commercial outcome is identical — the supplier is paid early, the buyer receives terms, the financier earns a return — but the legal reality is entirely different, and the financier's recovery position is stronger, because it holds title rather than a claim.

Making that real at scale is an engineering problem, and it is the problem this platform exists to solve. The **Contract Sequencing Engine** (Section 3.5) enforces the order and reality of every leg through hard gates, records typed evidence of ownership and possession at each stage, and hash-chains the executed documents so that the sequence can be verified cryptographically rather than taken on trust. No override role exists. The Shariah Supervisory Board can audit any transaction by replaying it.

1.3 Solution shape

Layer	Provided by	Responsibility
Sanad	IOTA (build)	All Islamic contract structuring, leg sequencing, ownership ledger, evidence chain, profit computation, decisioning, limits and facilities, document orchestration, Shariah governance, national rail integration, all user surfaces.

Layer	Provided by	Responsibility
Tuum	Licensed platform	Party and customer master, accounts, general ledger, payment execution, instalment schedule holding, servicing operations. <i>Lending intelligence deliberately not used.</i>
Nutrient	Licensed platform, self-hosted in-Kingdom	Bilingual document generation, embedded document review, form handling, OCR, redaction, template-drift comparison, PAdES digital signature and long-term archival.

Tuum is deliberately demoted from system of intelligence to system of record. This is not a workaround for a vendor gap; it is the design that makes Sanad a sellable product. Clients two through ten will not all run Tuum, and banks in particular already own a core. Concentrating the Islamic contract logic in Sanad also concentrates the audit surface where the Board can inspect it.

1.4 Phase one

Distributor Finance under Murabaha, with a single anchor, in a single sector. Wasl purchases goods from the anchor and sells them to the anchor's distributors on deferred terms. The anchor is paid on delivery; the distributor receives credit; Wasl earns a disclosed markup.

This is chosen as the first product for four reasons: the anchor is commercially motivated to drive adoption and therefore supplies distribution at no cost; ownership evidence is clean because the anchor's own systems produce it; credit risk is often mitigated by anchor recourse over its own dealer network; and the structure is unanimously accepted, so no fatwa risk sits on the critical path to launch.

1.5 What is deliberately excluded

Excluded	Reason
Invoice discounting, receivables purchase, factoring, reverse factoring in its conventional form	<i>Bay' al-dayn</i> — sale of debt below face value. Not permitted under the AAOIFI and Gulf position.
Tawarruq and any cash-generating structure	Contested. Organised tawarruq was ruled impermissible by the OIC International Islamic Fiqh Academy. Excluded from Phase 1 entirely; admissible later only on an explicit ruling of the Board, with enforced conditions and a volume cap.
Any interest-bearing instrument in any wrapper	<i>Riba</i> . No interest field exists in the data model.
Conventional insurance	Takaful only.
Consumer and retail lending	Out of scope for Wasl. Later products on the Sanad platform.

1.6 Critical path

Three items determine whether this programme starts on time, and none of them are engineering:

1. **A named anchor.** Without a large corporate willing to run a pilot programme there is no Phase 1, only a platform with nothing flowing through it.
2. **Agenda time with both Shariah Supervisory Boards.** Both institutions hold sitting Boards, which removes a multi-month appointment from the critical path. What remains is calendar: Boards meet on a cycle, and placing a

new product on an agenda commonly takes four to eight weeks. *Requesting those slots is the most time-sensitive action in the programme and should not wait for the questions to be finalised.*

3. **Intellectual property ownership.** The client engagement is bespoke; the commercial objective is a product. If the client owns the deliverable, the productisation strategy is void. This must be settled in the contract before build begins.

2. Business Requirements Document

2.1 Purpose and scope

This section states what the business requires of the Wasl product, in terms that can be traced to design, build and acceptance test. It defines the product set, the actors, the processes, and the functional, Shariah, regulatory and non-functional requirements.

In scope: origination, structuring, decisioning, documentation, execution, disbursement instruction, and lifecycle-to-settlement of Shariah-compliant supply chain finance transactions for SME counterparties in the Kingdom of Saudi Arabia, delivered to a bank or a licensed finance company, with the platform designed from the outset for repeat deployment.

Out of scope: the general ledger, payment rail execution, core account servicing, collections operations beyond origination handover, and consumer lending. These are either Tuum's responsibility or a later phase.

2.2 Business context

Factor	Implication for Wasl
Vision 2030 and Monsha'at	SME contribution to GDP is a declared national target. SME financing has institutional, political and regulatory tailwind. Government programmes exist specifically to de-risk it.
ZATCA FATOORA e-invoicing	Structured, cryptographically stamped, authority-cleared B2B invoices. This is the single most consequential input to both underwriting and Shariah evidence. Wasl is designed around it, not merely integrated with it.
Kafalah guarantee programme	Government-backed SME financing guarantee materially improves risk appetite and unit economics. Structure of the guarantee must be reviewed by the Board (<i>kafala</i> with a fee is a live question).
SAMA Open Banking	Account information services give consented access to SME cash-flow data, confirming what ZATCA data indicates.
Islamic finance depth	The market expects Shariah compliance as a baseline, and institutional Shariah governance is mature. A product that is compliant by construction, and auditable by the Board without IT involvement, is a genuine differentiator.
Data residency regime	SAMA Cloud Computing Regulatory Rules and PDPL together require in-Kingdom hosting for this data. This is an architectural constraint, not a preference, and it drives the Nutrient deployment model.

2.3 Problem statement

For the SME. A small supplier or distributor holds a confirmed order or an approved invoice from a creditworthy buyer, and cannot convert it into working capital. Conventional lenders underwrite the SME's own balance sheet, which is thin, unaudited and often invisible. Payment terms of sixty to a hundred and twenty days transfer the financing burden down the chain to the party least able to bear it.

For the anchor corporate. Its distribution network is capital-constrained, which caps its own sales volume. Extending terms itself consumes its working capital and puts SME credit risk on its balance sheet.

For the financier. Supply chain finance is attractive — short tenor, self-liquidating, granular — but conventional programmes fail on fraud, duplicate financing and unverifiable SME data, and the Islamic institution faces an additional and more fundamental obstacle: the standard product is a debt purchase and is not permitted.

2.4 Business objectives and success measures

Ref	Objective	Measure at end of Phase 1
OBJ-01	Originate compliant SME supply chain finance at scale	One live anchor programme; distributor drawdowns flowing daily; zero Shariah incidents rated material.
OBJ-02	Make non-compliance structurally impossible	Board confirms, on inspection of live transactions, that no path exists to execute a sale before ownership and possession are evidenced.
OBJ-03	Onboard SMEs in minutes, not days	Median distributor onboarding, from invitation to first available limit, under 30 minutes with no manual intervention in the straight-through case.
OBJ-04	Pay the counterparty same-day	Median time from drawdown acceptance to settlement instruction under 2 hours during business hours.
OBJ-05	Eliminate invoice fraud and duplicate financing	100% of financed invoices validated against ZATCA; zero successful duplicate financings.
OBJ-06	Give the Board self-service audit	Board members replay any transaction end to end, with evidence and signature validation, without an IT request.
OBJ-07	Produce a reusable product, not a bespoke system	Zero client-specific identifiers in the core codebase, enforced in continuous integration. All client variation expressed as configuration or adapter.

2.5 Stakeholders and personas

Persona	Surface	Needs
Distributor / SME owner Primary user. Often mobile-only, Arabic-first, low patience for process.	Wasl SME portal (web + mobile)	Know their available limit; request finance against a real order in a few taps; understand the cost as a fixed amount, not a rate; be paid or have their supplier paid quickly; see what they owe and when.
Anchor programme manager Treasury or commercial function at the large corporate.	Anchor portal	Onboard and invite their network; see utilisation against the programme limit; approve or confirm trades; reconcile settlements; report internally.
Credit officer / underwriter	Operations workbench	Work exceptions, not straight-through cases; see the decision trace and the evidence; set and review limits; exercise four-eyes approval.
Operations officer	Operations workbench	Resolve document, evidence and matching exceptions; manage supplier and goods registers; handle settlement failures.

Persona	Surface	Needs
Shariah Supervisory Board member / Shariah auditor	Shariah audit workspace	Sample any transaction; replay the leg sequence with timestamps; verify evidence and signatures; confirm the template version and its fatwa reference; export an audit pack. Without asking IT.
Internal Shariah reviewer	Shariah governance module	Monitor automated compliance checks; triage and remediate Shariah incidents; run purification.
Compliance / AML officer	Operations workbench	Screening alerts, PDPL data subject requests, regulatory reporting.
Risk / portfolio manager	Analytics	Concentration by anchor, sector and counterparty; vintage performance; early warning.
Platform administrator	Admin console	Tenant configuration, product parameters, template versions, users and entitlements.

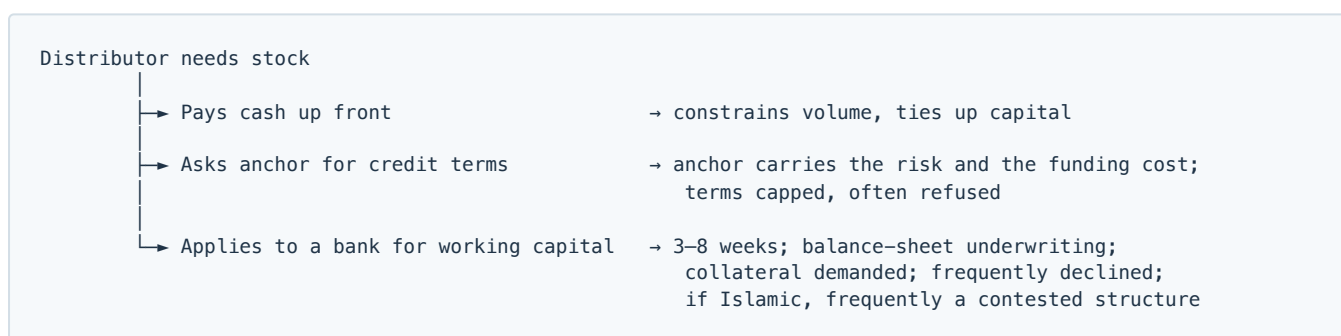
2.6 Product scope — the Wasl product set

Ref	Product	Islamic structure	Phase	Description
PR-01	Distributor Finance	Murabaha	PHASE 1	Wasl purchases goods from the anchor, takes title, and sells to the distributor on deferred payment at cost plus a disclosed profit. Anchor paid on delivery.
PR-02	Supplier Early Payment	Murabaha	PHASE 2	Wasl purchases goods from the SME supplier for immediate cash and sells them to the anchor buyer on deferred terms. The compliant equivalent of approved-payables finance.
PR-03	Purchase Order Finance — fungibles	Salam + Parallel Salam	PHASE 2	Wasl pays in advance for future delivery of specified fungible goods, and contracts a parallel onward sale. Serves producers with a confirmed order and no working capital.
PR-04	Purchase Order Finance — manufactured	Istisna'a + Parallel Istisna'a	PHASE 2	Wasl commissions manufacture against milestones and contracts an onward sale. Serves manufacturers and construction supply chains.
PR-05	Inventory Finance	Murabaha on stock, or Ijarah on warehousing	PHASE 3	Seasonal stocking cycles and held inventory.
PR-06	Equipment and Fleet	Ijarah Muntahia Bittamleek	PHASE 3	Capital equipment inside the supply chain, on a lease ending in ownership transfer.

Ref	Product	Islamic structure	Phase	Description
PR-07	Government Contract Finance	Istisna'a / Salam	PHASE 3	Financing SMEs that hold awarded government contracts, verified through Etimad. Government is the Kingdom's largest buyer.
PR-08	Growth Capital	Musharaka / Mudaraba	PHASE 3	True profit-and-loss partnership for selected SMEs within a chain.
PR-X1	Invoice discounting / receivables purchase	—	EXCLUDED	<i>Bay' al-dayn</i> . Not permitted. No implementation path exists in the platform.
PR-X2	Tawarruq / cash finance	—	EXCLUDED	Contested. Excluded from Phase 1. Admissible only on explicit Board ruling with enforced conditions and volume caps.

2.7 Business process — as-is and to-be

2.7.1 As-is: the distributor's current experience



2.7.2 To-be: the Wasl transaction

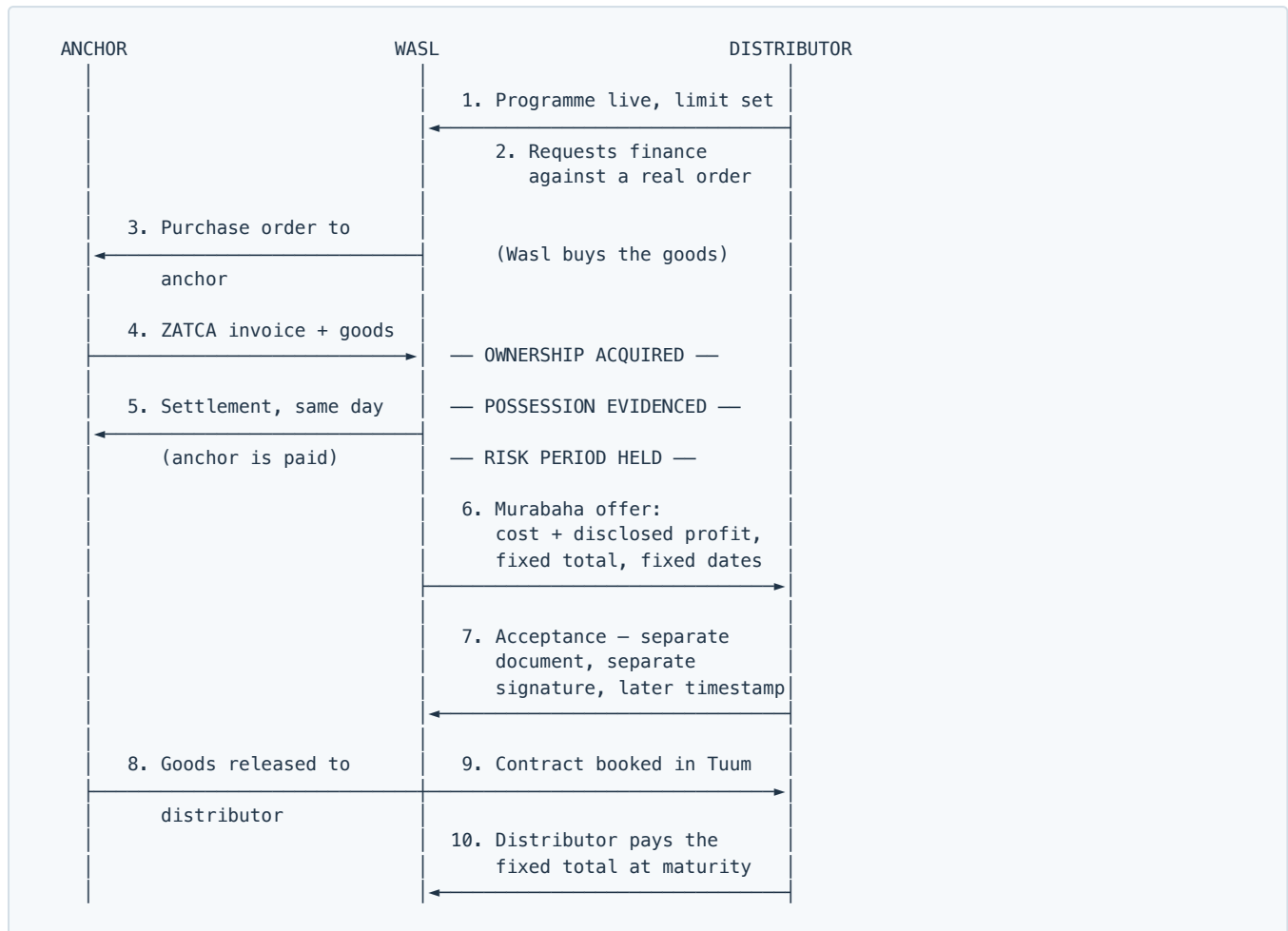


Figure 2.1 — Distributor Finance under Murabaha. Steps 4, 5 and the risk period are hard gates: step 6 is unreachable until all three are satisfied and evidenced.

The commercial effect matches conventional distributor finance exactly. The legal substance does not: Wasl owns goods and sells them, rather than lending money against paper.

2.8 Functional requirements

2.8.1 Anchor and programme management

Ref	Requirement	Priority
BR-A01	The system shall onboard an anchor corporate, capturing legal entity, commercial registration, authorised signatories, sectors, and the goods categories to be financed.	Must
BR-A02	The system shall support definition of a <i>programme</i> as the commercial container for an anchor relationship: product, structure, currency, tenor range, profit rate basis, programme limit, eligible counterparty criteria, goods register, and recourse terms.	Must
BR-A03	A programme shall not be activatable unless it is linked to an in-force Shariah approval record covering its product structure and its contract template set.	Must
BR-A04	The system shall maintain a programme limit, allocate sub-limits to counterparties within it, and prevent aggregate counterparty utilisation from exceeding the programme limit.	Must

Ref	Requirement	Priority
BR-A05	The anchor shall be able to invite counterparties into a programme in bulk, by upload or by API, and track their onboarding status.	Must
BR-A06	The anchor shall be able to view programme utilisation, transaction status, settlement position and overdue exposure in near real time.	Must
BR-A07	The system shall support anchor recourse and buyback terms as programme configuration, and reflect them in limit and risk calculations.	Should
BR-A08	The system shall support multiple concurrent programmes per anchor, and the same counterparty participating in programmes of different anchors, with correctly aggregated exposure.	Must

2.8.2 Counterparty onboarding

Ref	Requirement	Priority
BR-B01	The system shall onboard an SME counterparty using verified national sources, minimising manual data entry: commercial registration and ownership from the Ministry of Commerce business data service, authorised signatory identity through Nafath, national address, and workforce data.	Must
BR-B02	The system shall verify that the individual completing onboarding is an authorised signatory of the legal entity, and record the verification evidence.	Must
BR-B03	The system shall screen the counterparty, its owners and its authorised signatories against sanctions and politically exposed person lists, and rescreen on a configurable schedule and on material change.	Must
BR-B04	The system shall obtain and record explicit, granular, withdrawable consent for each category of data access, satisfying PDPL, with a machine-readable consent record.	Must
BR-B05	The system shall retrieve the counterparty's e-invoicing history and commercial credit report, subject to consent, as primary underwriting inputs.	Must
BR-B06	The system shall retrieve bank account information through Open Banking, subject to consent, as a confirmatory underwriting input.	Should
BR-B07	The system shall complete straight-through onboarding, from invitation to an assigned limit, without manual intervention where all verifications pass and no policy exception is raised.	Must
BR-B08	The system shall route any failed verification, screening hit or policy exception to a named operations queue with the full evidence pack attached.	Must
BR-B09	The system shall screen the counterparty's business activity for permissibility against a Board-approved excluded-activity register, and record the outcome with reasons.	Must
BR-B10	The system shall re-verify commercial registration validity, signatory authority and screening status before each drawdown, using cached data within a configurable freshness window.	Must

2.8.3 Limit assignment and decisioning

Ref	Requirement	Priority
BR-C01	The system shall assign a counterparty limit by applying a versioned, Board- and Credit-approved policy to verified data, primarily the counterparty's e-invoicing trade history with the anchor.	Must
BR-C02	Every decision shall be reproducible: given the same input snapshot and the same policy version, the system shall return an identical outcome.	Must
BR-C03	Every decision shall persist the complete input snapshot, the policy version, the full rule trace, the score, and human-readable reason codes.	Must

Ref	Requirement	Priority
BR-C04	Decline and limit-reduction reasons shall be issuable to the counterparty in Arabic, derived from the rule trace rather than authored by hand.	Must
BR-C05	The system shall support champion-challenger policy execution, where the challenger evaluates without affecting the outcome.	Should
BR-C06	The system shall recompute limits on a schedule and on trigger events, and shall support limit increase, decrease, suspension and withdrawal with reasons and audit.	Must
BR-C07	The system shall enforce concentration controls at counterparty, anchor, sector and goods-category level, and block drawdowns that would breach them.	Must
BR-C08	Credit policy shall be maintained as versioned configuration with effective dating, authored and approved outside the release cycle, and never encoded in application code.	Must

2.8.4 Transaction origination and execution

Ref	Requirement	Priority
BR-D01	A counterparty shall be able to initiate a drawdown against a specified real trade — a purchase order or an invoice — never against a cash amount alone.	Must
BR-D02	The system shall validate the referenced invoice against the national e-invoicing authority, confirming existence, issuer, recipient, amount, date and cryptographic integrity.	Must
BR-D03	The system shall maintain a registry of financed invoice identifiers and shall reject any drawdown referencing an identifier already financed, in whole or in part.	Must
BR-D04	The system shall verify that every line item falls within the programme's permitted goods register, and shall block the transaction where any item is on the excluded-goods register.	Must
BR-D05	The system shall verify that the ultimate buyer and the original seller in a transaction are distinct legal entities, matched on registration identifier rather than name.	Must
BR-D06	The system shall perform a three-way match between purchase order, invoice and delivery evidence, with configurable tolerances, and raise an exception on mismatch.	Must
BR-D07	The system shall compute the sale price as cost plus a profit amount fixed at inception, and shall present cost and profit separately and explicitly to the counterparty.	Must
BR-D08	The system shall generate each contractual leg as a separate bilingual document, executed separately, timestamped separately, and cryptographically chained in sequence.	Must
BR-D09	The system shall not permit the onward sale leg to be offered or executed until ownership and possession of the goods are recorded with valid evidence and the configured minimum risk-holding period has elapsed.	Must
BR-D10	No user role, including administrative roles, shall be able to override the sequencing gates in BR-D09.	Must
BR-D11	The system shall instruct settlement to the selling counterparty on the same business day as ownership transfer, where the payment rail permits.	Must
BR-D12	On execution of the sale leg, the system shall book the resulting obligation in the core banking platform exactly once, with idempotent retry and automated reconciliation.	Must
BR-D13	The system shall support partial drawdown against an approved trade, and multiple concurrent drawdowns under one counterparty limit.	Should
BR-D14	The system shall support transaction cancellation and unwind prior to sale-leg execution, with documented disposition of any goods already acquired.	Must

2.8.5 Lifecycle, collections and hardship

Ref	Requirement	Priority
BR-E01	The total amount owed by a counterparty shall be fixed at execution and shall not increase under any subsequent event, including default, rescheduling or extension.	Must
BR-E02	The system shall support rescheduling that changes instalment dates and amounts only, and shall reject any rescheduling whose instalment sum differs from the original total.	Must
BR-E03	Late payment amounts shall post exclusively to a segregated charity liability account. No accounting path shall exist to recognise them as income.	Must
BR-E04	Late payment amounts shall apply only where the counterparty is assessed as able but unwilling to pay. An assessment of inability shall route the case to the hardship process.	Must
BR-E05	The system shall provide a first-class hardship process: assessment, evidence capture, deferral, rescheduling without price increase, profit waiver where the Board permits, and a documented write-off path.	Must
BR-E06	The system shall apply early settlement relief in exactly the form ruled by the Board, reconciled with the applicable regulatory requirement, as configuration rather than code.	Must
BR-E07	The system shall hand over delinquent cases to the servicing platform's collections function with the full origination evidence pack.	Must
BR-E08	The system shall track and report goods-level recovery rights where title-based security is retained.	Should

2.8.6 Shariah governance

Ref	Requirement	Priority
BR-F01	The system shall hold a Shariah approval record for each product structure and template set, capturing the approving body, the approval reference, the approving members, the effective date and the expiry or review date.	Must
BR-F02	Expiry, suspension or withdrawal of a Shariah approval shall automatically suspend new origination on every affected product, without manual action.	Must
BR-F03	The system shall run continuous automated compliance checks on live transactions covering sequence integrity, risk-period compliance, evidence completeness, counterparty distinctness, goods permissibility and disclosure accuracy.	Must
BR-F04	Any check failure shall raise a Shariah incident with severity, owner, remediation plan, and mandatory closure evidence.	Must
BR-F05	The system shall provide a read-only Shariah audit workspace permitting sampling, full transaction replay with timestamps and evidence, signature validation, template version confirmation and audit pack export, without technical assistance.	Must
BR-F06	The system shall compute non-compliant income arising from defective transactions or incidental impermissible receipts, segregate it automatically, and support disbursement to Board-approved recipients with a complete trail.	Must
BR-F07	The system shall compare every executed document against its approved template version and raise a Shariah incident on any deviation.	Must
BR-F08	The system shall produce periodic Shariah compliance reporting suitable for submission to the Board and inclusion in the annual Shariah report.	Must

2.8.7 Documents and signature

Ref	Requirement	Priority
BR-G01	All contractual documents shall be produced bilingually, with Arabic as the governing text and a precedence clause stating so.	Must
BR-G02	Templates shall separate Board-approved immutable clauses from commercially variable fields. The rendering engine shall be incapable of altering a locked clause.	Must
BR-G03	Every generated document shall carry, in visible form and in metadata, its template identifier, template version and the Shariah approval reference under which it was executed.	Must
BR-G04	Master agreements shall be executed with a digital signature carrying long-term validation, using a certificate issued by a licensed national certification service provider.	Must
BR-G05	Per-transaction legs shall be executed by electronic acceptance bound to a verified identity assertion, sealed with the institution's own certificate, and hash-chained to the preceding leg and to the master agreement.	Must
BR-G06	The system shall capture and retain a complete signing evidence record: identity assertion, timestamp from a trusted time source, device and network context, document hash, and the exact rendition presented.	Must
BR-G07	The system shall generate documents asynchronously at volume without degrading interactive response times.	Must
BR-G08	The system shall extract structured data from uploaded unstructured evidence — delivery notes, warehouse receipts, transport documents — by optical character recognition, with confidence scoring and exception routing.	Should
BR-G09	The system shall redact documents automatically and by rule before disclosure to third parties such as guarantors, takaful operators or external counsel.	Must
BR-G10	Executed documents shall be archived in a long-term preservation format with signature validation material embedded, for the full statutory retention period.	Must

2.9 Shariah requirements

These are not policies to be followed. They are properties the system must possess. Each is stated so that its absence is testable.

Ref	Principle	System obligation
SH-01	No riba	No interest rate, accrual or compounding construct shall exist in the domain model or the persistence layer. Return is expressed exclusively as a profit <i>amount</i> , derived once at inception as sale price less cost, and immutable thereafter.
SH-02	No riba on default	No code path shall exist by which an executed obligation's total increases after execution. Rescheduling APIs shall reject any payload whose instalment sum differs from the original total.
SH-03	No gharar	Goods, price, payment dates and delivery terms shall be fully determinate before a contract can reach an executable state. Any null in the determinacy set shall block execution.
SH-04	No maysir	No contingent pricing, optionality or derivative feature shall be offered.
SH-05	No sale before ownership	The transaction state machine shall be structurally incapable of reaching the sale leg before ownership and possession are recorded with valid typed evidence.

Ref	Principle	System obligation
SH-06	Genuine ownership risk	A minimum risk-holding interval, set by the Board per product, shall be enforced by the system clock between possession and sale. It shall not be waivable by any role.
SH-07	Separation of contracts	Each leg shall be a distinct document with its own execution event and timestamp. The document service shall refuse to render two legs into a single instrument.
SH-08	No 'inah	Counterparty distinctness shall be enforced on verified registration identifiers across the full transaction chain, including indirect paths.
SH-09	No sale of debt	Receivables shall be marked non-transferable other than at par. No API, report or export shall support discounted assignment of a receivable.
SH-10	No duplicate financing	A financed invoice identifier shall be permanently registered and shall not be financeable again. This is a Shariah control as well as a credit control: financing the same goods twice means selling what was not owned.
SH-11	Permissible goods only	Line-item level screening against a Board-maintained excluded-goods register shall block the transaction, not warn on it. The institution must never take title to impermissible goods.
SH-12	Permissible counterparty activity	Business activity screening against Board-set criteria shall be a gate at onboarding and a periodic review thereafter.
SH-13	Late charges are not income	Late payment amounts shall post to a segregated charity liability account. The chart of accounts mapping shall make income recognition impossible, not merely discouraged.
SH-14	Forbearance for genuine hardship	A hardship path shall exist as a first-class process, not an exception. Establishing inability to pay shall suspend late charges automatically.
SH-15	Disclosure of cost and profit	Original cost and profit markup shall be disclosed explicitly in the sale document and in the user interface. In Murabaha this is a validity condition, not a consumer-protection courtesy.
SH-16	Takaful only	No conventional insurance product shall be offerable or referenceable. Provider registers shall contain licensed takaful operators only.
SH-17	Fatwa-bound operation	No product shall be activatable without a live Shariah approval record. Approval lapse shall automatically suspend origination.
SH-18	Auditability by the Board	The Board shall be able to verify any of the above on any transaction, by itself, without technical assistance.

DESIGN NOTE

Requirements SH-01 through SH-11 are deliberately expressed as *absences* — of a field, of a code path, of an API, of an override. This is the difference between a system that is compliant and a system that cannot be made non-compliant. Acceptance testing for these requirements is adversarial: the test attempts to produce the prohibited outcome and passes only when it cannot.

2.10 Regulatory and compliance requirements

Ref	Regime	Obligation on the system
RC-01	SAMA licensing	The platform shall operate within the licence held by the deploying institution. Licence scope, permitted activities and reporting obligations shall be confirmed before build and reflected in product configuration.
RC-02	SAMA Cyber Security Framework	Controls mapped, implemented and evidenced. Control-to-implementation traceability maintained as a delivery artefact.
RC-03	SAMA Cloud Computing Regulatory Rules	Data classified; hosting in-Kingdom for all classifications requiring it; documented exit plan; approvals obtained where required.
RC-04	SAMA business continuity	Recovery objectives defined and tested; continuity plan exercised; degraded-mode operation defined per external dependency.
RC-05	PDPL (SDAIA)	Lawful basis recorded per processing purpose; granular consent capture and withdrawal; data subject access, correction and erasure; cross-border transfer controls; breach notification workflow; documented retention schedule.
RC-06	AML / CFT	Counterparty, beneficial owner and signatory screening; ongoing monitoring; transaction monitoring for trade-based money laundering typologies; suspicious transaction reporting workflow.
RC-07	ZATCA e-invoicing	Integration conformant with the published technical specification; correct handling of the invoice XML, its cryptographic stamp and its unique identifier; correct treatment of VAT on the institution's own purchase and sale legs.
RC-08	VAT	The institution buys and sells goods. VAT treatment of both legs shall be modelled correctly and confirmed with tax advisors. This is a material difference from a lending product and must not be assumed away.
RC-09	Shariah governance standards	Three-line governance implemented: Board approval, internal Shariah review, independent Shariah audit — each with distinct entitlements and segregated duties.
RC-10	Electronic transactions law	Signature patterns shall satisfy the statutory conditions for enforceability. Certificates shall chain to a licensed national certification service provider. Legal opinion obtained before Phase 1 exit.
RC-11	Consumer and SME conduct	Mandatory disclosures rendered; complaint capture and routing; clear, Arabic, non-misleading presentation of cost.
RC-12	Audit and records	Immutable audit trail of every state transition, decision and document event; records retained for the full statutory period in a form that remains verifiable.

ATTENTION — RC-08

Because Wasl takes title to goods, the institution is a party to two taxable supplies per transaction rather than a lender making an exempt or out-of-scope financing. VAT treatment, input recovery and invoicing obligations differ materially from a conventional lending product. This must be resolved with tax counsel in Phase 0; it affects pricing, the chart of accounts, and the ZATCA integration design.

2.11 Non-functional requirements

Ref	Attribute	Target
NFR-01	Interactive latency	95th percentile under 500 ms for read operations and under 1.5 s for transactional writes, measured at the API boundary, excluding third-party calls.

Ref	Attribute	Target
NFR-02	Throughput	Design point: 5,000 drawdowns per day per tenant at launch, with headroom to 50,000 without re-architecture. Supply chain finance is a high-volume, low-value profile; this inverts the usual lending assumption.
NFR-03	Availability	99.9% for counterparty-facing services during business hours; 99.5% for internal surfaces. Degraded mode defined for every external dependency.
NFR-04	Recovery objectives	RPO 15 minutes; RTO 4 hours. Both tested, not asserted.
NFR-05	Data residency	All customer, transaction and document data stored and processed in-Kingdom. No exception, including for logs, backups, analytics and third-party processors.
NFR-06	Document generation	Asynchronous; 10,000 documents per day sustained; interactive surfaces never blocked on generation.
NFR-07	Language	Arabic-first. Full right-to-left support across every surface. Complete parity of content between Arabic and English.
NFR-08	Calendar	Hijri and Gregorian supported throughout, with user preference, and correct handling in schedules, documents and reporting.
NFR-09	Accessibility	WCAG 2.2 level AA on all surfaces, verified in both language directions.
NFR-10	Mobile	SME surfaces fully functional on a mid-range Android device over a constrained mobile network. This is the primary access mode for the primary persona.
NFR-11	Auditability	Every state transition, decision and document event recorded immutably with actor, timestamp, before and after state, and correlation identifier.
NFR-12	Multi-tenancy	Tenant isolation enforced at the data layer. Single-tenant deployment topology; multi-tenant codebase. No client-specific identifier in core code, verified in continuous integration.
NFR-13	Configurability	Products, policies, templates, screening registers, workflows, entitlements and branding shall be configuration. A new client shall require no core code change.
NFR-14	Portability	The core banking platform shall be replaceable by implementing one adapter. No core banking vendor concept shall appear in the Sanad domain model.
NFR-15	Observability	Distributed tracing across all services and external calls; business-level metrics alongside technical; every transaction traceable end to end by correlation identifier.

2.12 Assumptions, dependencies and constraints

Type	Statement
Assumption	A named anchor corporate will be committed before Phase 1 begins, and will adapt its procurement or distribution process to accommodate the financier taking title.
Assumption	The deploying institution holds, or will hold, a SAMA licence permitting these activities, and funds the book from its own balance sheet or a committed line.
Confirmed	Both target institutions — an Islamic bank and a fintech — hold sitting Shariah Supervisory Boards. No Board appointment sits on the critical path. The residual constraint is Board agenda availability, not Board existence.
Assumption	The two Boards may rule differently on contested points. The platform accommodates divergence through per-tenant configuration (Section 3.11) rather than by seeking a common position.
Assumption	Each institution's existing Shariah governance artefacts — fatwas, governance manual, approved template set, review methodology and report formats — will be made available in Phase 0. These materially shorten the design cycle.

Type	Statement
Assumption	Constructive possession, properly evidenced, will be accepted by the Board. If physical possession is required, the operating model changes materially.
Dependency	Technical and commercial access to ZATCA e-invoicing data, the Ministry of Commerce business data service, Nafath, the commercial credit bureau, Open Banking and the payment rails.
Dependency	Tuum sandbox access and written confirmation of Islamic-finance and supply-chain capability.
Dependency	Nutrient self-hosted licensing for in-Kingdom deployment.
Dependency	Selection and contracting of a licensed national certification service provider for digital signatures. This has the longest lead time of any dependency.
Constraint	All data in-Kingdom. This constrains cloud region, managed service selection and third-party processor choice.
Constraint	Arabic is the governing contractual language.
Constraint	No structure outside the Board-approved set may be implemented, however commercially attractive.
Constraint	Intellectual property in the core platform must remain with IOTA for the productisation strategy to be viable.

2.13 Business risks

Ref	Risk	Impact	Mitigation
RSK-01	No anchor committed	Critical	Anchor commitment is a Phase 0 exit gate. Do not begin Phase 1 without a signed letter of intent.
RSK-02	Anchor unwilling to change procurement so the financier can take title	Critical	Test this in the first anchor conversation, not after design. Distributor Finance is chosen partly because the anchor is the <i>seller</i> , which is a smaller operational change than the payables-side variant.
RSK-03	A Board rejects constructive possession or requires longer risk-holding than the economics support	High	Resolve in Phase 0 before design freeze. Model the economics at several risk-period settings, and put the question to both Boards in parallel.
RSK-13	The two Boards diverge on a contested point and the divergence is discovered after build	High	Run the parallel divergence exercise in Phase 0 (Section 3.11). Every identified divergence point must be configuration, verified by test. A divergence found in build is a redesign; found in Phase 0 it is a configuration key.
RSK-14	Board agenda lead time delays Phase 0 exit	Medium	Request agenda slots with both Boards immediately, ahead of finalising the question set. Calendar cannot be compressed retrospectively.
RSK-04	Core banking platform cannot support Islamic accounting or supply-chain volume	High	Six written questions in week one. Fallback architecture already defined: Sanad owns contract logic and the facility engine; the core holds accounts, ledger and payments only.
RSK-05	Core banking commercial model priced per contract, breaking unit economics at supply-chain volume	High	Obtain volume pricing before Phase 1 pricing is presented to the client.
RSK-06	National rail onboarding is calendar-driven and slower than the build	High	Begin commercial and technical onboarding on day one, in parallel with design.
RSK-07	Arabic document rendering fidelity insufficient for a governing-language contract	High	Phase 0 spike against the most complex real template, reviewed by Legal.

Ref	Risk	Impact	Mitigation
RSK-08	Digital signature pattern not accepted as enforceable	High	Certification provider selected in Phase 0; a validated signed document produced before design freeze; legal opinion before Phase 1 exit.
RSK-09	VAT treatment of the two supply legs erodes margin	High	Tax counsel engaged in Phase 0. Pricing model built on the confirmed treatment.
RSK-10	Trade-based money laundering through the programme	Medium	Authority-cleared invoices, three-way matching, goods screening, counterparty distinctness and transaction monitoring together make this substantially harder than in a conventional programme.
RSK-11	Bespoke build never productises	High	IP retained; multi-tenant model from first commit; three-layer separation enforced; second-client test in code review; client on released versions, never on the main branch.
RSK-12	Anchor concentration — programme dominated by one counterparty relationship	Medium	Concentration limits enforced in the platform; multi-anchor capability delivered in Phase 2.

2.14 Scope exclusions

The following are explicitly not delivered by Wasl and are either the responsibility of the core banking platform, a later phase, or another system entirely: general ledger and financial accounting; payment rail connectivity and clearing; customer and account servicing operations; collections operations beyond the origination handover; treasury and funding management; regulatory financial reporting; consumer and retail lending of any kind; and any form of invoice discounting, receivables purchase or cash financing.

3. High Level Design

3.1 Solution overview

Sanad is an origination platform. Wasl is a product configuration on it. The platform provides Islamic contract structuring, sequencing and evidence; decisioning; limits and facilities; document orchestration; Shariah governance; national rail integration; and the user surfaces. It delegates accounts, ledger and payments to a core banking platform behind an adapter, and document rendering and signature to a document platform behind a service boundary.

Three design commitments shape everything that follows:

1. **Compliance is structural.** The Shariah requirements in Section 2.9 are enforced by the state machine and the data model, not by validation that a privileged user could bypass.
2. **Evidence is first-class.** Every claim the institution would need to defend before its Board — that it owned the goods, that it held them, that the documents were separate and sequential — is captured as typed, timestamped, tamper-evident evidence at the moment it becomes true.
3. **The core is replaceable.** Nothing in the Sanad domain model knows the name of a core banking vendor.

3.2 System context

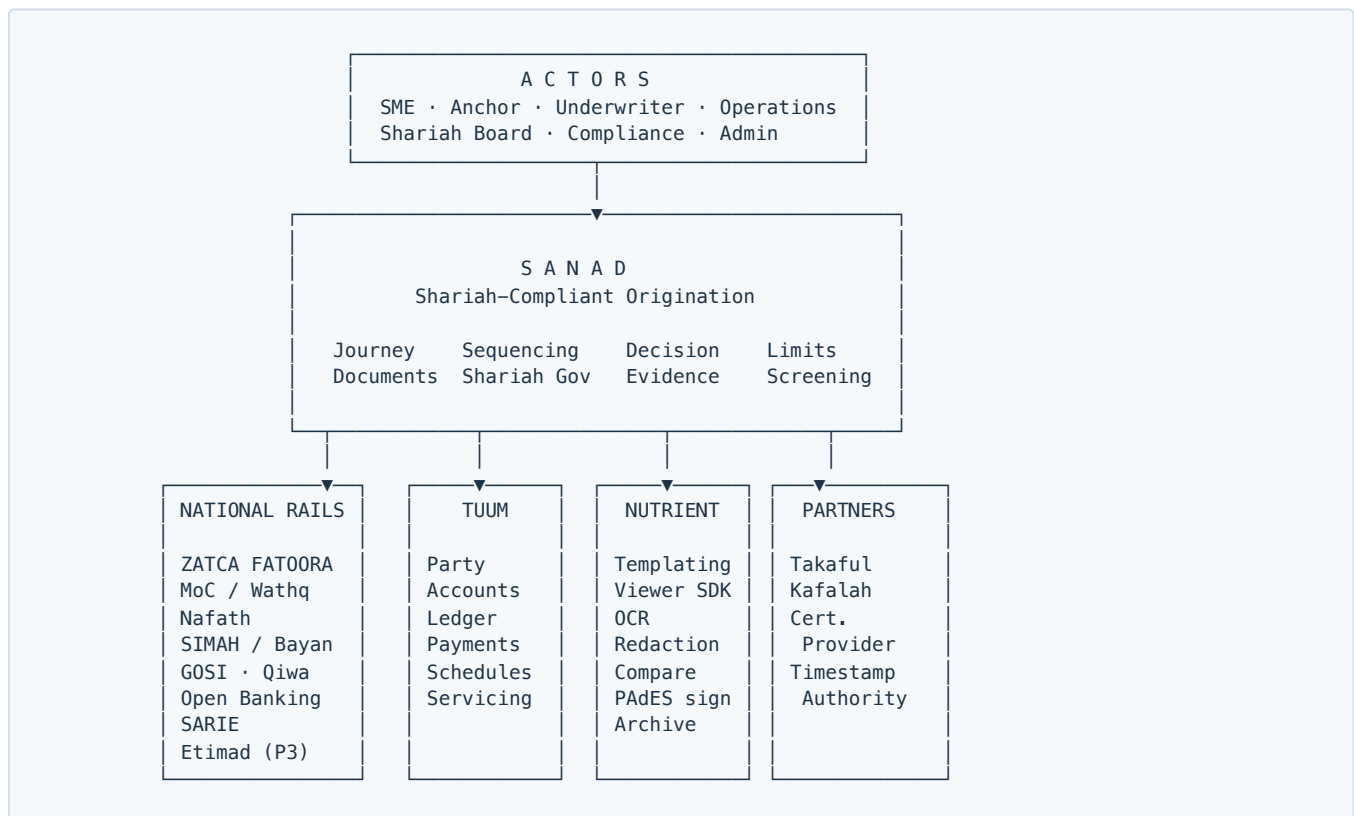


Figure 3.1 — System context. All components deployed in-Kingdom, including the document platform.

3.3 Capability model

Capability	Owner	Description
Anchor & programme management	Sanad	Anchor onboarding, programme definition, programme limits, counterparty invitation, utilisation reporting.
Counterparty onboarding	Sanad	Verified registry-sourced onboarding, signatory verification, screening, consent capture.
Decisioning	Sanad	Policy-as-code evaluation, scorecards, limit assignment, reproducible decision records.
Limit & facility management	Sanad	Programme, counterparty and concentration limits; real-time utilisation under concurrency.
Contract sequencing	Sanad	Leg orchestration, hard gates, ownership and possession evidence, risk-period enforcement, hash chaining.
Trade validation	Sanad	Invoice authenticity, duplicate registry, goods screening, counterparty distinctness, three-way match.
Document orchestration	Sanad → Nutrient	Template selection and version binding, bilingual rendering, batch generation, signature orchestration, archival, drift comparison.
Shariah governance	Sanad	Approval records, continuous review, incident management, audit workspace, purification.
Settlement instruction	Sanad → Tuum	Disbursement and collection instruction, reconciliation.
Party, accounts, ledger	Tuum	Customer master, account opening, postings, balances.
Payment execution	Tuum	Rail connectivity, settlement, returns.
Servicing & collections	Tuum	Instalment collection, delinquency, operational servicing.
Document rendering & signature	Nutrient	Generation, viewing, form handling, OCR, redaction, comparison, PAdES signing, long-term archive.

3.4 The Shariah problem and the structural resolution

Stated plainly so that the engineering consequence is unambiguous:

Conventional SCF instrument	Shariah analysis	Wasl equivalent
Reverse factoring / approved payables finance — financier buys the supplier's approved receivable at a discount	<i>Bay' al-dayn</i> . Sale of debt below face value. Not permitted.	Supplier Early Payment Murabaha . Financier buys the <i>goods</i> from the supplier for cash and sells them to the anchor on deferred terms.
Invoice discounting / receivables finance	<i>Bay' al-dayn</i> . Not permitted.	No equivalent. Excluded.
Distributor finance as a working capital loan	<i>Riba</i> . Not permitted.	Distributor Finance Murabaha . Financier buys goods from the anchor and sells them to the distributor on deferred terms.
Pre-shipment / purchase order loan	<i>Riba</i> . Not permitted.	Salam for fungibles, Istisna'a for manufactured goods, with a parallel onward contract.

Conventional SCF instrument	Shariah analysis	Wasl equivalent
Invoice discounting restyled as a "service fee" on a discounted receivable	<i>Hiyal</i> — a ruse. The substance is unchanged. Not permitted.	Explicitly rejected as a design option.

THE ENGINEERING CONSEQUENCE

Because the financier buys and sells real goods, Wasl is not a lending system with a compliance layer. It is a **trading system** that produces a financing outcome. It must model goods, ownership, possession, title, delivery, VAT on two supplies, and physical or constructive custody. A conventional lending data model cannot be adapted to this; the domain model in Section 5 is built for it from the start.

The compensating benefit is material: because the institution holds title rather than a claim, its position on counterparty default is stronger than the conventional lender's, not weaker.

3.5 Contract Sequencing Engine

This is the core of the platform and the component on which the Board's assurance rests.

3.5.1 Principle

In Islamic commercial law the validity of a Murabaha depends on facts about the world in a particular order: the institution acquired the goods, took possession of them, held them at its own risk, and only then sold them. A Murabaha where the institution never truly owned the goods is a loan wearing a costume. The origination system is precisely where that is either made real or quietly lost.

3.5.2 Enforced sequence — Murabaha

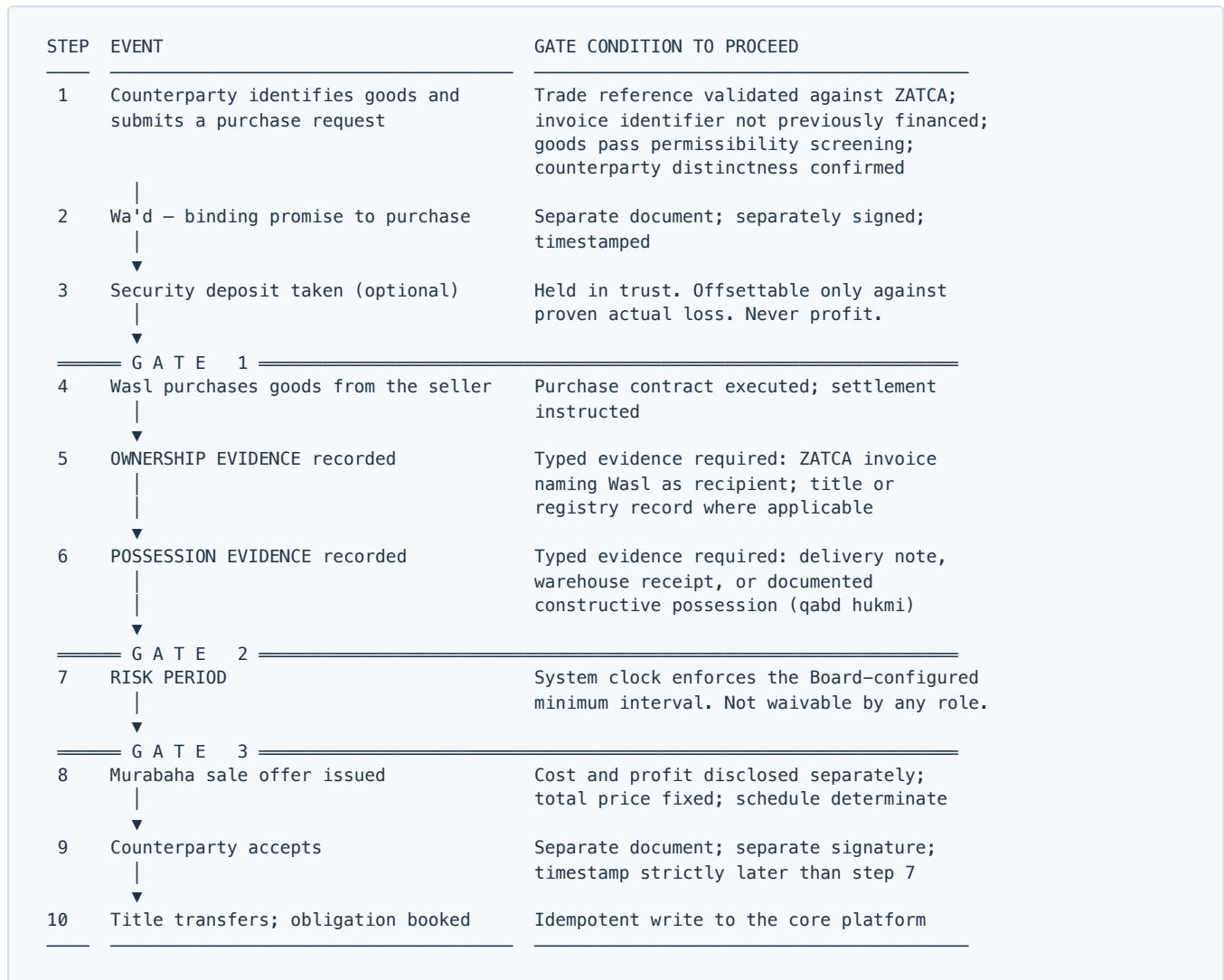


Figure 3.2 — Murabaha leg sequence. Steps 8 onward are unreachable in the state machine until Gates 1, 2 and 3 are all satisfied.

3.5.3 How the gates are made unypassable

Mechanism	Effect
State machine, not validation	The transition to the sale-offer state is not defined in the state machine unless the prerequisite states are occupied. There is no code path to skip it, so there is nothing for an override to override.
No override role	No entitlement exists — for underwriters, operations, administrators or support — that permits gate bypass. This is a deliberate absence, tested adversarially.
Typed evidence	Each gate requires evidence of a declared type with declared validation. A checkbox is not evidence. A document of the wrong type does not satisfy the gate.
Clock-enforced risk period	The interval between possession and sale is enforced against a trusted time source, not against a user-supplied timestamp.
Cryptographic sequence proof	Each leg's executed document is timestamped by a trusted timestamping authority and hash-chained to its predecessor. The order is verifiable mathematically, independent of the application's own logs.
Continuous review	Automated Shariah review re-verifies sequence integrity on live transactions and raises an incident on any anomaly, independently of the transaction path itself.

3.5.4 Other structures

The same engine, differently configured, enforces the sequences for Salam (price paid in full at contract; goods specified; delivery date determinate; parallel contract independent of the first), Istisna'a (specification determinacy; milestone-linked payment; parallel contract independent), and Ijarah (institution holds ownership throughout the lease term, bears ownership risk and major maintenance, and transfers title only by a separate instrument at the end). Each structure is a declarative sequence definition validated against the Board's approval record, not bespoke code.

3.6 Master agreement and per-transaction execution

Shariah rigour requires that each leg be a separate contract, separately executed. Supply chain finance volume makes it impossible for a counterparty's authorised signatory to apply a cryptographic signature to hundreds of documents a month. The resolution is a two-tier execution model.

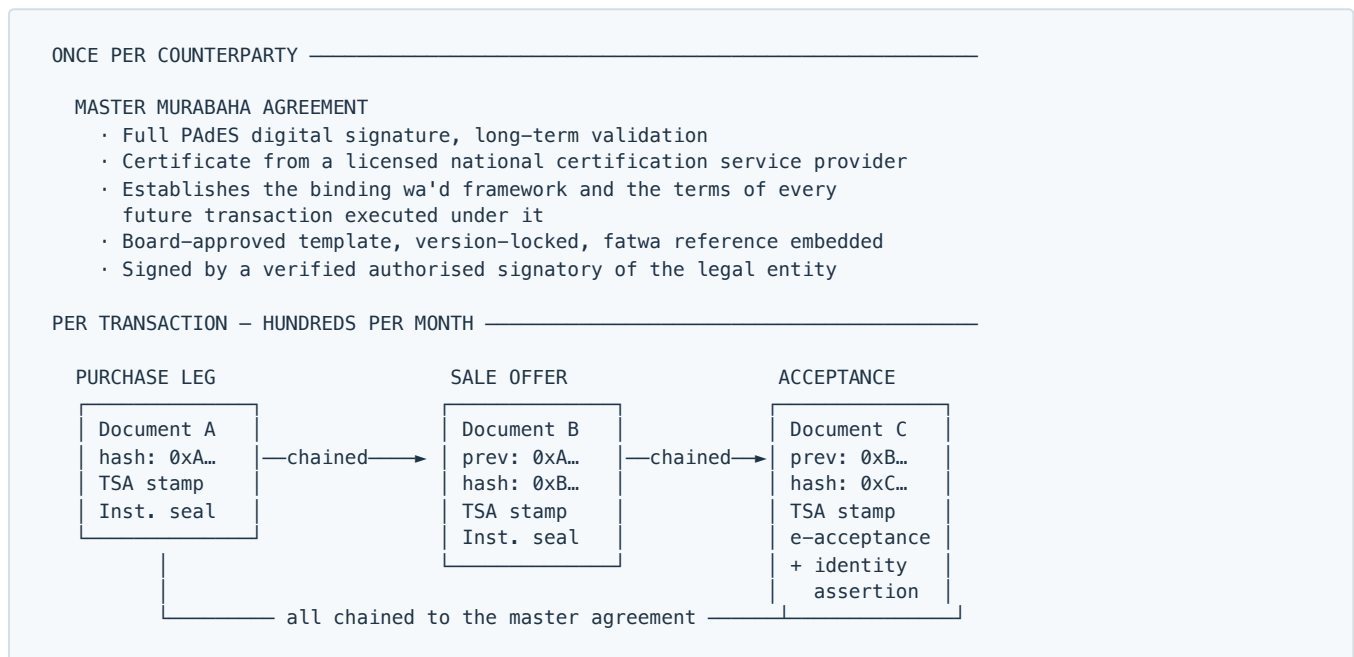


Figure 3.3 — Two-tier execution. Legs remain separate instruments; only the master agreement requires full ceremonial signing.

BOARD RULING REQUIRED — OI-04

This pattern is the single design decision most likely to attract Board scrutiny, and it must be ruled on explicitly in Phase 0. The question to put to the Board is whether electronic acceptance under a pre-executed master agreement, bound to a verified identity assertion and sealed with the institution's certificate, constitutes valid execution of a separate contract. If the Board requires full ceremonial signing of every leg, the product's operating model and its unit economics both change materially, and Phase 1 scope must be revisited.

3.7 End-to-end journeys

3.7.1 Anchor programme establishment

Commercial agreement → Anchor entity onboarded and verified → Programme defined (product, structure, limits, tenor, pricing basis, goods register, recourse)

- Shariah approval record linked and verified in force
- Master agreement executed with the anchor (full digital signature)
- Programme activated → Counterparty invitation list loaded

Deliberately heavyweight and human. Measured in weeks. This is correct — the programme is where the credit appetite, the Shariah structure and the commercial terms are set.

3.7.2 Counterparty onboarding — target under thirty minutes

Invitation (SMS / email, Arabic)

- Authorised signatory authenticates via Nafath
- Commercial registration retrieved and verified from the national business register
- Signatory authority confirmed against the registration record
- Granular consent captured (e-invoicing, credit bureau, open banking, workforce)
- Screening: sanctions, PEP, adverse media
- Business activity permissibility screening
- Trade history retrieved from the e-invoicing authority
- Credit bureau and open banking data retrieved
- Decision engine evaluates → limit assigned
- Master agreement rendered bilingually and digitally signed
- Counterparty active, limit available

ANY failure, hit or exception → operations queue with the full evidence pack.

Never a dead end for the user; always a clear status and an expected response time.

3.7.3 Drawdown — the high-volume path

Counterparty selects a real trade (purchase order or ZATCA invoice)

- └ Invoice validated with the e-invoicing authority
 - └ Invoice identifier checked against the financed registry
 - └ Goods screened line by line against the permitted register
 - └ Counterparty distinctness verified on registration identifiers
 - └ Limit availability checked and reserved
 - └ Three-way match: purchase order ↔ invoice ↔ delivery evidence
- } all must pass

SEQUENCING ENGINE – Gates 1, 2, 3 as per Figure 3.2

- └ Purchase leg executed; settlement instructed to the seller
- └ Ownership and possession evidence captured
- └ Risk period observed
- └ Sale offer issued: cost + profit disclosed, total fixed
- └ Acceptance captured

Obligation booked in the core platform (idempotent)

Documents sealed, chained and archived

Limit utilisation confirmed

Events published to analytics and Shariah review

3.7.4 Settlement and maturity

At maturity the counterparty pays the fixed total. Payment is collected by the core platform and reconciled by Sanad against the transaction. Partial payments reduce the outstanding balance and never change the total. On delinquency, Sanad determines whether the case is able-but-unwilling — in which case a late amount accrues to the charity liability account — or a genuine hardship, in which case the hardship process opens and late charges are suspended automatically.

3.8 ZATCA e-invoicing as the system of truth

Three failure modes historically destroy supply chain finance programmes. Saudi national infrastructure substantially resolves all three, and Wasl is designed to exploit that rather than merely to interoperate with it.

Classic failure	Resolution in Wasl
Fabricated invoices	Invoices are generated in structured form, cryptographically stamped and cleared with the tax authority. Wasl validates the stamp and the clearance rather than trusting a submitted PDF. Fabrication ceases to be a realistic attack.
Duplicate financing	Every invoice carries a unique identifier. Wasl maintains a permanent registry of financed identifiers. A second financing attempt against the same identifier, at this institution, is structurally blocked. <i>Cross-institution duplicate financing remains an industry-level gap and is noted as OI-09.</i>
Unverifiable SME revenue	The counterparty's cleared invoice history is a tax-authority-verified record of who it sells to, how much, how often and on what terms. This becomes the primary underwriting input; bank statements and bureau data become confirmatory.

The Shariah dividend is equally large. A cleared invoice naming the institution as recipient, combined with a delivery record and a title transfer, is strong third-party evidence of a real trade — verifiable by the Board rather than attested by the institution about itself.

STRATEGIC NOTE

This design is portable. Every jurisdiction adopting mandatory structured e-invoicing — across the Gulf, and in Malaysia, Egypt and Türkiye among others — becomes a market where the same underwriting and evidence model works, requiring only a new adapter. The e-invoicing adapter interface should therefore be designed as a generic contract from the first commit, not as a ZATCA-specific integration.

3.9 Platform boundary — Sanad, Tuum and Nutrient

Decision	Rationale
Sanad owns all Islamic contract logic	Concentrates the audit surface where the Board can inspect it; removes dependency on a vendor's Islamic-finance roadmap; and keeps the platform core-agnostic, which is the precondition for selling it to clients that run a different core.
Tuum holds party, accounts, ledger, payments and schedules	These are genuine strengths and there is no reason to rebuild them. The lending intelligence is deliberately unused.
Sanad may also own the facility and limit engine	Contingent on OI-02. Supply chain finance requires revolving, multi-drawdown limits with real-time utilisation under concurrency, which term-lending modules commonly do not provide.
Nutrient self-hosted in-Kingdom	Contract documents contain national identifiers, commercial terms and signatures. Residency rules make a non-Kingdom document endpoint unacceptable. Self-hosting is available and is the reason this platform fits.

Concern	Sanad	Tuum	Nutrient
Islamic structures & legs	•	—	—
Sequencing & evidence	•	—	evidence sealing
Decisioning & limits	•	exposure feed	—
Profit computation	•	—	—

Concern	Sanad	Tuum	Nutrient
Party & account master	reference	•	—
Ledger & postings	—	•	—
Payment execution	instruction	•	—
Instalment schedule	computes	holds & collects	renders
Document generation	orchestrates	—	•
Signature & archive	orchestrates	—	•
Shariah governance	•	—	template control

3.10 Multi-tenancy and productisation model

The engagement is bespoke; the objective is a product. These commonly conflict, and the conflict is resolved by architecture rather than by intention.

Discipline	Implementation
Multi-tenant from the first commit	Tenant identifier on every row; tenant-scoped configuration resolution; no global singletons. Deployment remains single-tenant — banks require it — but the codebase never assumes one tenant.
Three-layer separation	<code>core</code> contains structures, sequencing, decisioning and document orchestration and holds no client-specific code. <code>adapters</code> contain every external system. <code>config</code> holds products, policies, templates, registers, workflows and branding as data.
The second-client test	A standing question in code review: would the next client want this exact behaviour? If not, it belongs in configuration or an adapter. Present in the pull request template.
No client identity in code	Enforced by a continuous integration check. It is trivial in itself and serves as the canary for every other leak of client specificity into the core.
Client on released versions	The first client runs a tagged release, never the main branch. The moment production is patched directly, there is one bespoke system and no product.
Two packages, one codebase	<i>Sanad Core</i> — the origination layer for institutions that already own a core banking platform. <i>Sanad Complete</i> — bundled with Tuum, hosted, for fintech clients wanting a turnkey stack.

3.11 Multi-board Shariah configuration

Two institutions are in scope, each with its own sitting Shariah Supervisory Board. **They may rule differently on the same question, and both rulings will be correct for their own institution.** This is normal in Islamic finance and it is not a problem to be resolved by persuasion. It is a design requirement.

3.11.1 Where divergence is likely

Contested point	Ref	Consequence if the platform assumes one position
Validity of electronic acceptance under a pre-executed master agreement	OI-04	Severe. The product is unusable for the institution whose Board requires full ceremonial execution of every leg. Operating model and unit economics both change.

Contested point	Ref	Consequence if the platform assumes one position
Acceptability of constructive possession, and what evidence discharges <i>qabd</i>	OI-07	Different evidence types required at Gate 2; possibly a different operating model if physical possession is demanded.
Minimum risk-holding period	OI-07	Directly changes the economics and the settlement cycle.
<i>ibra'</i> formulation on early settlement	OI-12	Different calculation; different reconciliation with the regulatory requirement.
Late payment basis and charity routing method	OI-12	Different accounting treatment and different disclosure.
Purification methodology and approved recipients	OI-12	Different computation and disbursement process.
Goods and business activity screening thresholds	OI-15	Different registers and different decline populations.
Treatment of a fee-bearing government guarantee	OI-16	One institution may use the guarantee programme and the other may not.

3.11.2 Why the existing design already accommodates this

Three decisions taken for other reasons turn out to carry the whole multi-board requirement:

- **Structure sequences are declarative data** (Section 6.5.1), not code. A different leg order, a different gate definition or a different evidence requirement is a different structure definition, not a different build.
- **Shariah parameters are versioned, effective-dated configuration** (Section 5.7) — risk-holding periods, registers, the *ibra'* basis, charity treatment and purification methodology are all held as tenant-scoped data owned by the Board that set them.
- **Every transaction snapshots its governing approval** (Section 5.4.1). A transaction records the `shariah_approval_id` and the parameters in force when it executed, so each Board audits its own institution's transactions against its own rulings — including historic ones, after a parameter has since changed.

THE BINDING RULE

Anything a Board can rule on is configuration. Nothing a Board can rule on is code. If a divergence between the two Boards would require a code change, the design is wrong at that point and must be parameterised before build. This is the test to apply to every Shariah-relevant decision during Phase 1.

3.11.3 The two failure modes to avoid

Failure mode	Why it fails
Build to the union — satisfy the strictest position on every point	Produces a product neither institution wants. The strictest position on every axis simultaneously is commercially unusable, and no Board asked for it.
Build to the intersection — implement only what both Boards accept	Produces a product neither Board approves, because each has requirements the other does not share and both are mandatory for their own institution.

Failure mode	Why it fails
Build to client #1 and adapt later	The most common failure and the most expensive. Client #1's fatwa becomes embedded in code and in the test suite, and is discovered only during client #2's Board review — by which point it is a redesign, not a configuration change.

The correct approach is parameterisation with per-tenant approval binding, which is what the platform does.

3.11.4 The parallel divergence exercise

A Phase 0 activity, roughly two weeks of effort, with disproportionate value:

1. Put **the same question set to both Boards in parallel**, rather than sequentially.
2. Where the Boards agree, adopt the agreed position as the **platform default**.
3. Where they differ, that point becomes a **named configuration key**, specified before any code is written.
4. Produce a **divergence map** recording each question, each Board's position, and the configuration key that absorbs the difference. This becomes a delivery artefact and a test input.
5. **Parameterise the adversarial compliance suite** (Section 6.13) against the divergence map, so the tests run per tenant configuration rather than encoding one Board's rulings.

COMMERCIAL NOTE

The divergence map is also a sales asset. A documented account of where credible Saudi Boards differ, together with demonstrated evidence that the platform absorbs those differences by configuration, is a difficult thing for a competitor to answer — and it is the most concrete proof available that Sanad is a product rather than one institution's bespoke system.

4. Architecture

4.1 Architecture principles

Ref	Principle	Consequence
AP-01	Compliance by construction	Shariah and regulatory invariants are enforced by types, state machines and schema constraints — never by validation that a privileged path could skip.
AP-02	Evidence at the moment of truth	Evidence is captured when a fact becomes true, not reconstructed later from logs. Reconstruction is not evidence.
AP-03	One system of record per fact	Sanad owns the transaction and the evidence chain; the core platform owns balances and postings. Neither duplicates the other, and reconciliation proves it nightly.
AP-04	Vendor concepts stop at the adapter	No vendor DTO, identifier scheme or terminology enters the domain model. Replacing a vendor is an adapter implementation.
AP-05	Configuration over code	Products, policies, templates, registers and workflows are versioned, effective-dated data authored outside the release cycle.
AP-06	Idempotent by default	Every externally visible write accepts an idempotency key. Retries are safe. Duplicate financial effects are not possible.
AP-07	Fail closed on compliance, fail soft on convenience	An unavailable compliance dependency blocks the transaction. An unavailable enrichment dependency degrades the experience without blocking it.
AP-08	Asynchronous where the user is not waiting	Document generation, notification, enrichment, reporting and archival are queued. Interactive paths stay short.
AP-09	In-Kingdom, without exception	Region selection, managed service selection and processor selection are all constrained by residency, including for logs, backups and analytics.
AP-10	Arabic is not a locale bolted on	Right-to-left, bilingual content and Hijri dates are modelled in the data and the design system from the first commit.

4.2 Layered architecture

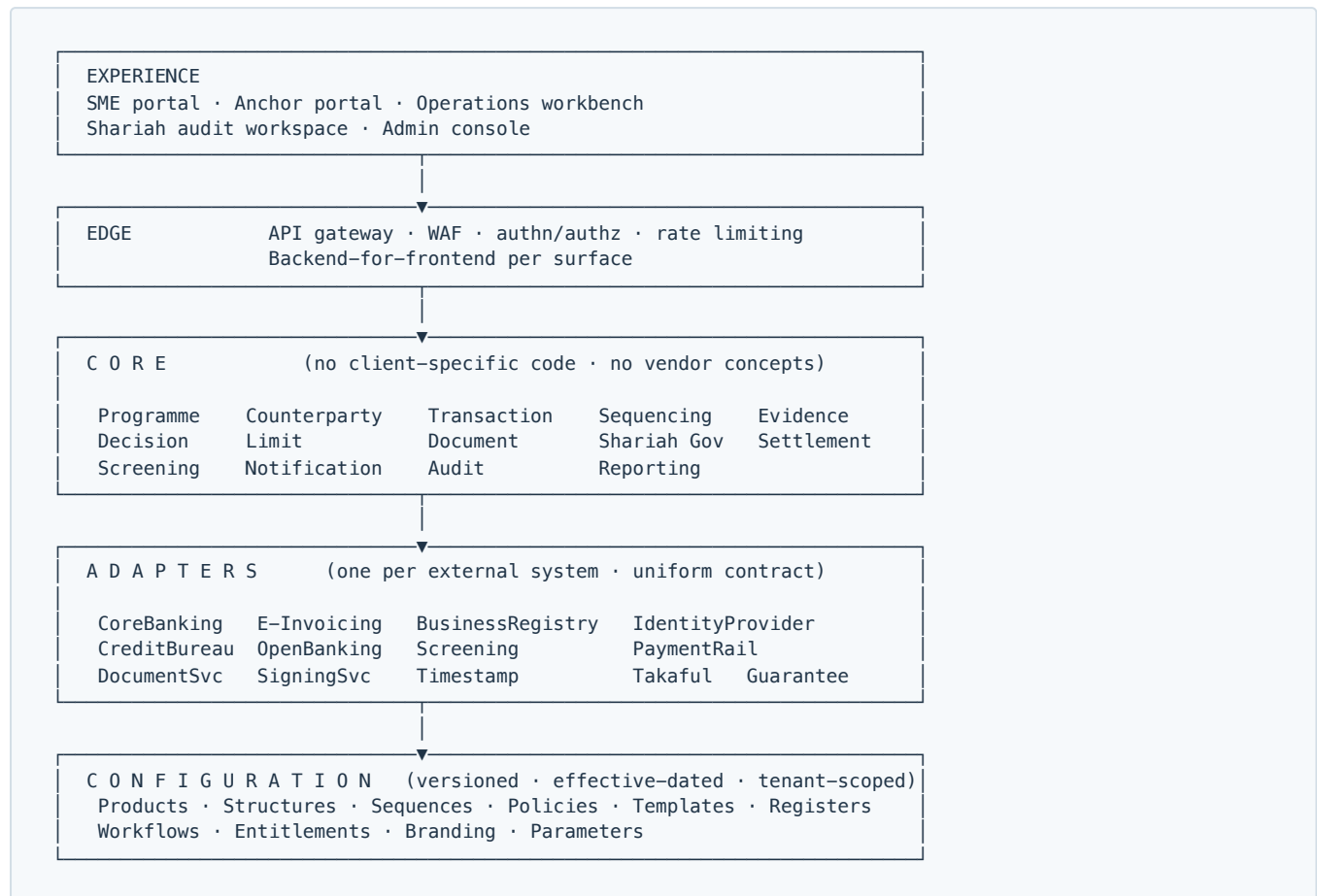


Figure 4.1 — Layer separation. Client-specific work lands in adapters and configuration; the core is what IOTA sells repeatedly.

4.3 Logical architecture

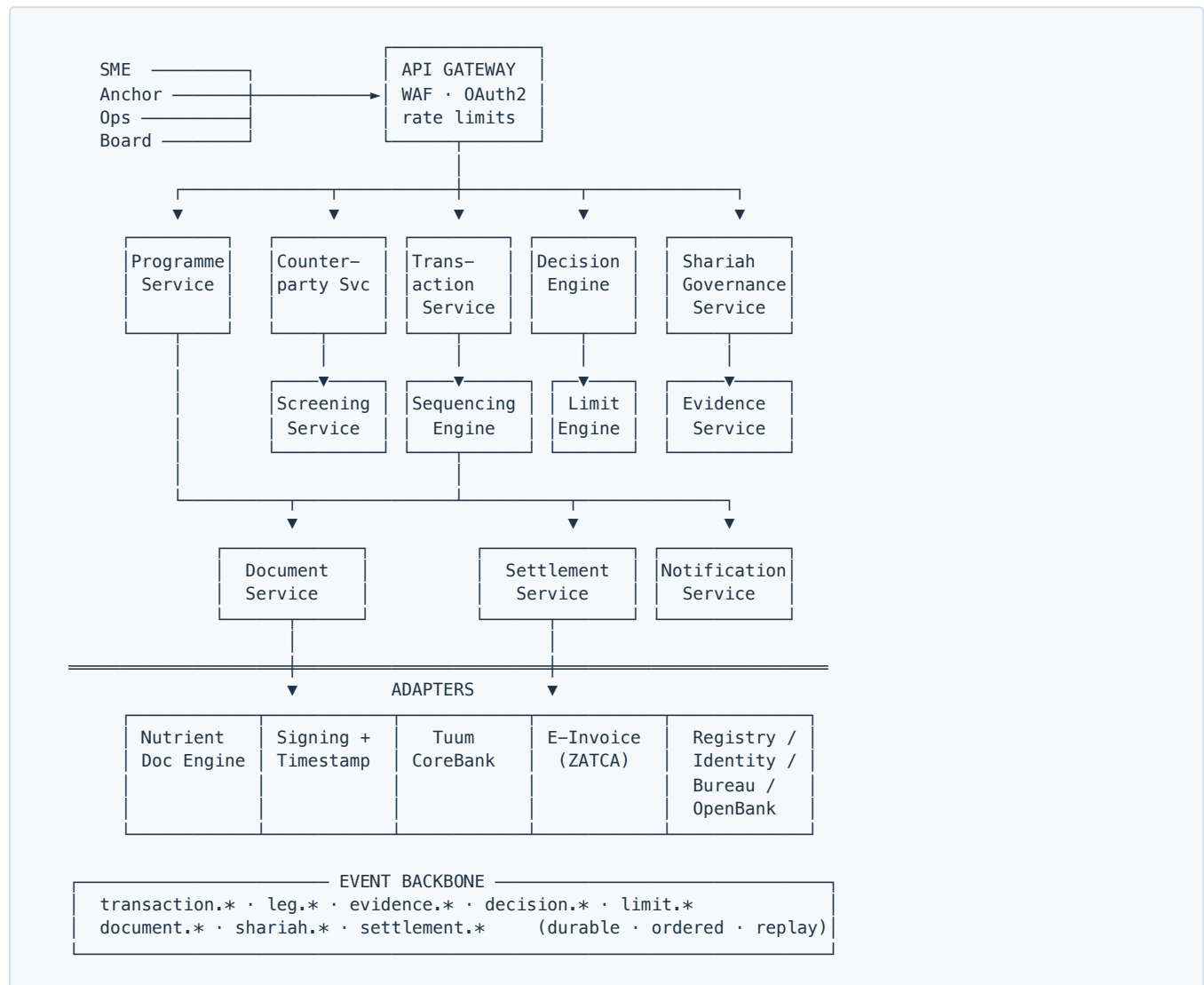


Figure 4.2 — Logical architecture. Services communicate synchronously for request-scoped work and via the event backbone for everything else.

4.4 Service decomposition

Service	Owns	Notes
Programme Service	Anchors, programmes, programme limits, counterparty invitations	Low volume, high configuration. Guards activation on Shariah approval.
Counterparty Service	SME entities, signatories, consents, onboarding state, verification records	Orchestrates registry, identity, bureau and open banking adapters.
Screening Service	Sanctions, PEP, adverse media, business activity permissibility, goods permissibility	Registers are configuration maintained by Compliance and the Board.
Decision Engine	Policy evaluation, scorecards, decision records, reason codes	Deterministic and reproducible. Policy is versioned configuration.

Service	Owns	Notes
Limit Engine	Programme, counterparty and concentration limits; reservation and utilisation	The highest-concurrency component. Reservation is the correctness-critical operation.
Transaction Service	The drawdown aggregate, trade validation, financed-invoice registry, three-way match	Highest volume. The system's hot path.
Sequencing Engine	Structure definitions, leg orchestration, gates, risk-period enforcement, hash chaining	The compliance heart. Structure definitions are declarative and bound to Shariah approvals.
Evidence Service	Typed evidence capture, validation, immutable storage, integrity verification	Append-only. Nothing deletes evidence.
Document Service	Template binding, generation orchestration, signature orchestration, archival, drift comparison	Fronts Nutrient. Batch and asynchronous by default.
Settlement Service	Disbursement and collection instruction, reconciliation, returns handling	Fronts the core banking adapter. Outbox-driven and idempotent.
Shariah Governance Service	Approval records, continuous review, incidents, purification, audit workspace data	Reads across every service. Writes nothing to transactions except incident links.
Notification Service	Arabic-first SMS, email and push; delivery tracking	Templated; bilingual; delivery evidence retained.
Audit Service	Immutable event log, correlation, export	Append-only, hash-chained, independently verifiable.

4.5 Integration architecture

4.5.1 Interaction patterns

Pattern	Used for	Behaviour on failure
Synchronous request / response	Invoice validation, identity verification, registry lookup, limit reservation	Fail closed. The transaction cannot proceed. User sees a clear status and a retry path.
Synchronous with cache	Registry data, bureau data, programme configuration	Serve from cache within a configured freshness window; beyond it, fail closed.
Asynchronous command with outbox	Core banking booking, settlement instruction	Persisted in a transactional outbox, retried with backoff, reconciled nightly. Never lost, never duplicated.
Asynchronous job	Document generation, OCR, notification, archival	Queued with retry and dead-letter. Interactive path unaffected.

Pattern	Used for	Behaviour on failure
Event publication	Downstream analytics, Shariah review, anchor reporting	At-least-once with ordered partitions; consumers idempotent.
Event consumption	Core banking lifecycle events	Idempotent handlers; replay supported from the log.

4.5.2 External integrations

Ref	System	Pattern	Failure	Purpose
INT-01	E-invoicing authority (ZATCA)	Sync	Fail closed	Invoice authenticity, integrity, trade history
INT-02	Business registry (MoC / Wathq)	Sync + cache	Fail closed	Commercial registration, ownership, signatories
INT-03	Identity provider (Nafath)	Sync	Fail closed	Signatory authentication and assertion
INT-04	Commercial credit bureau	Sync + cache	Degrade	Credit history, existing obligations
INT-05	Open banking	Async	Degrade	Confirmatory cash-flow data
INT-06	Screening provider	Sync	Fail closed	Sanctions, PEP, adverse media
INT-07	Core banking (Tuum)	Async outbox + events	Queue & reconcile	Party, accounts, booking, payments, schedules
INT-08	Document platform (Nutrient)	Async job	Queue	Generation, OCR, redaction, comparison, archive
INT-09	Signing service / certification provider	Sync callback	Fail closed	PADES signature, institutional seal
INT-10	Timestamping authority	Sync	Fail closed	Trusted time on every leg
INT-11	Payment rail	Via core banking	Queue & reconcile	Same-day settlement
INT-12	Workforce data (GOSI / Qiwa)	Sync + cache	Degrade	Business substance signals
INT-13	Guarantee programme (Kafalah)	Async	Degrade	Guarantee application and status — Phase 2
INT-14	Takaful operator	Async	Degrade	Trade credit takaful — Phase 2
INT-15	Government procurement (Etimad)	Sync	Fail closed	Contract award verification — Phase 3

ADAPTER CONTRACT DISCIPLINE

Each adapter implements a generic, capability-named interface — `EInvoicingProvider` , `BusinessRegistryProvider` , `CoreBankingProvider` — and never exposes the vendor's vocabulary upward. Every adapter ships with recorded fixtures for deterministic testing, a circuit breaker, a defined freshness window, and a declared failure posture. This is what makes both the e-invoicing model portable to other jurisdictions and the core banking platform replaceable.

4.6 Event catalogue

Event	Published by	Principal consumers
<code>counterparty.onboarded</code>	Counterparty Service	Limit Engine, Notification, Analytics
<code>counterparty.screening.hit</code>	Screening Service	Operations, Compliance, Audit
<code>decision.rendered</code>	Decision Engine	Limit Engine, Analytics, Audit
<code>limit.assigned</code> / <code>limit.changed</code>	Limit Engine	SME portal, Anchor portal, Analytics
<code>transaction.initiated</code>	Transaction Service	Sequencing Engine, Analytics
<code>trade.validated</code> / <code>trade.rejected</code>	Transaction Service	Sequencing Engine, Operations, Shariah Review
<code>leg.executed</code>	Sequencing Engine	Document Service, Evidence Service, Shariah Review, Audit
<code>evidence.captured</code>	Evidence Service	Sequencing Engine, Shariah Review, Audit
<code>gate.passed</code> / <code>gate.blocked</code>	Sequencing Engine	Shariah Review, Operations, Audit
<code>ownership.acquired</code> / <code>possession.confirmed</code>	Sequencing Engine	Shariah Review, Analytics, Audit
<code>contract.executed</code>	Sequencing Engine	Settlement Service, Core Banking adapter, Analytics
<code>document.generated</code> / <code>document.sealed</code>	Document Service	Evidence Service, Audit, Notification
<code>document.drift.detected</code>	Document Service	Shariah Governance — raises an incident
<code>settlement.instructed</code> / <code>settlement.confirmed</code>	Settlement Service	Transaction Service, Anchor portal, Analytics
<code>shariah.incident.raised</code>	Shariah Governance	Operations, Board reporting, Purification
<code>purification.computed</code> / <code>purification.disbursed</code>	Shariah Governance	Finance, Audit, Board reporting
<code>hardship.opened</code>	Transaction Service	Servicing, Late-charge suppression, Analytics

4.7 Security architecture

Domain	Design
Identity — external users	National identity provider for signatory authentication, with the assertion retained as evidence. Session binding to the verified legal entity and role.
Identity — internal users	Enterprise single sign-on with mandatory multi-factor authentication. No local accounts for staff.
Authorisation	Role-based entitlements with tenant, programme and data-scope qualification, evaluated centrally and enforced at the service boundary. No entitlement grants sequencing-gate bypass.
Segregation of duties	Enforced between credit decisioning, operations, Shariah review and Shariah audit. The audit role is read-only by construction, not by convention.
Data in transit	TLS 1.3 throughout, including service-to-service. Mutual TLS for external partner integrations.
Data at rest	Full encryption; column-level encryption for national identifiers and financial detail; keys in a hardware security module with documented rotation.
Signing keys	The institutional sealing certificate resides in an HSM. Private key material is never handled by application code. Remote signing for counterparty certificates via the certification provider's interface.
Secrets	Centrally managed, dynamically issued, short-lived. No secret in source control, configuration files or container images.
Network	Segmented by trust zone: public edge, application, data, and a restricted zone for signing and key material. Egress allow-listed.
Application security	Dependency and container scanning, static and dynamic analysis in the pipeline, annual independent penetration testing, secure development lifecycle mapped to the SAMA framework.
Audit	Append-only, hash-chained audit log with independent verification. Administrative actions logged to a separate, differently-permissioned store.
Privacy	Data minimisation at collection; purpose limitation enforced by consent scope; pseudonymisation in non-production; no production data in lower environments under any circumstances.

4.8 Deployment and infrastructure

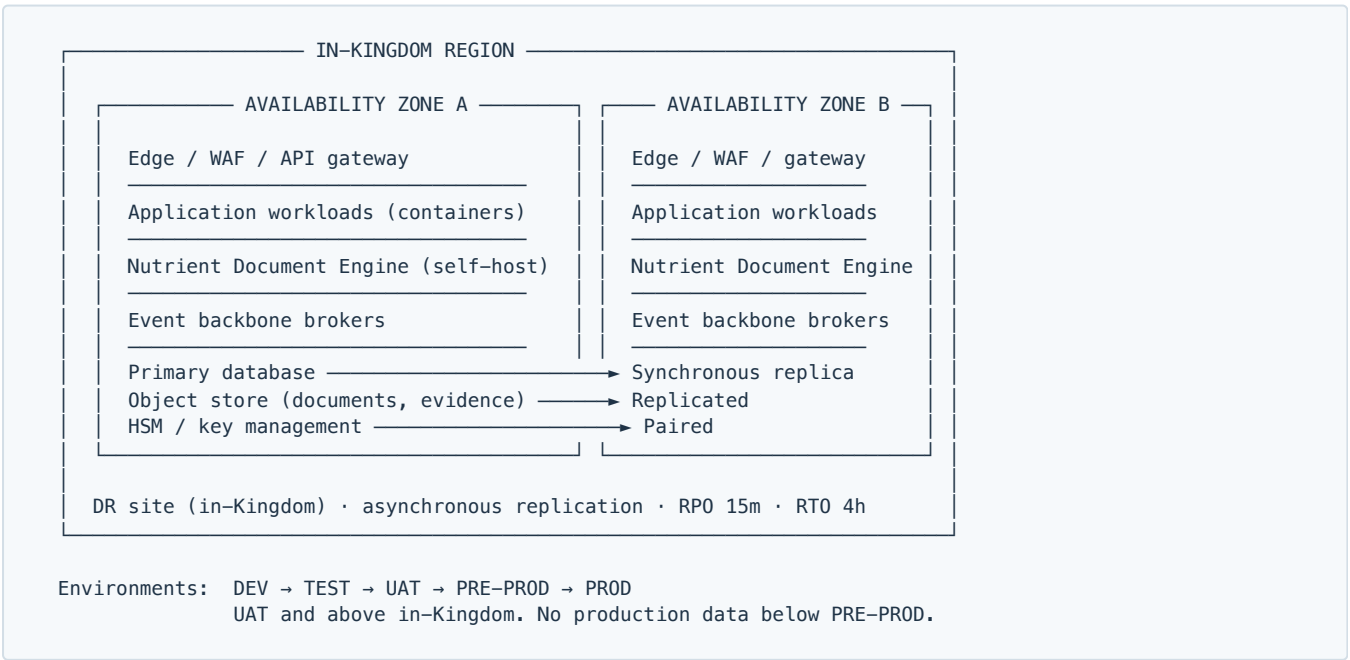


Figure 4.3 — Deployment topology. Single-tenant deployment per client institution; multi-tenant codebase.

Deployment is container-based on a managed orchestration platform, provisioned entirely as code, with immutable images and progressive rollout. Every environment is reproducible from source. The document platform is deployed inside the same trust boundary as the application, not consumed as an external service, which is the requirement that makes the residency position defensible.

4.9 Resilience and degraded modes

National rails go down. The design question is not whether but what happens then. Every external dependency has a declared posture, and the user-facing behaviour is specified rather than emergent.

Dependency unavailable	Posture	System behaviour
E-invoicing authority	Fail closed	New drawdowns blocked. Existing transactions continue. Clear Arabic status message with expected resumption. Queued for automatic retry.
Identity provider	Fail closed	New onboarding blocked; existing authenticated sessions continue.
Business registry	Cache, then fail closed	Serve cached registration data within the freshness window; beyond it, route to manual verification queue rather than blocking outright.
Credit bureau	Degrade	Onboarding continues to a reduced provisional limit, flagged for review when data becomes available.
Open banking	Degrade	Confirmatory input only; absence does not block.
Screening provider	Fail closed	No counterparty activation and no drawdown without a current screening result. Non-negotiable.
Core banking platform	Queue & reconcile	Origination continues; bookings and settlement instructions accumulate in the outbox and drain on recovery. Reconciliation confirms exactly-once.
Document platform	Queue	Transactions proceed to the point requiring a document, then hold. Generation resumes automatically. Users see an honest pending state.
Signing or timestamping service	Fail closed	No leg executes without a trusted timestamp and a valid seal. Compliance dependencies never degrade.
Payment rail	Queue & reconcile	Settlement instructions queue; the seller is notified of the delay; nothing is lost or duplicated.

Circuit breakers isolate a failing dependency before it exhausts connection pools. Bulkheads prevent one integration's latency from degrading unrelated paths. Every failure surfaces to the user as an Arabic status with a realistic expectation, never as a technical error.

4.10 Observability

Layer	Content
Tracing	Distributed traces spanning every service and every external call, correlated by a transaction identifier that is visible in the user interface so support can find any case instantly.
Technical metrics	Latency distributions, error rates, saturation and queue depth per service; external dependency latency and availability tracked independently.
Business metrics	Drawdowns initiated and completed, straight-through rate, time to settle, limit utilisation, exception queue depth and age, onboarding funnel conversion.

Layer	Content
Compliance metrics	Gate block counts by reason, evidence completeness, risk-period distribution, Shariah incidents by severity and age, template drift detections, purification balance. These are dashboards for the Board, not only for engineering.
Alerting	Tiered by consequence. A Shariah incident alerts differently, and to different people, than a latency breach.
Log handling	Structured, in-Kingdom, retained per the retention schedule, with national identifiers and financial detail redacted at source.

4.11 Technology selections

Concern	Selection	Rationale
Backend runtime	JVM (Kotlin or Java) with Spring Boot, or .NET — aligned to the client's existing operational capability	Both are proven in Saudi banking, have mature HSM, cryptography and PDF libraries, and are supportable by locally available talent. Selection should follow the client's operating model, not architectural preference.
Primary datastore	PostgreSQL	Strong transactional guarantees for the sequencing invariants, good JSON support for evidence payloads and policy documents, row-level security for tenant isolation, and available as a managed in-Kingdom service.
Event backbone	Apache Kafka	Durable, ordered, replayable. Replay matters: Shariah review and analytics must be able to rebuild from the log.
Cache	Redis	Freshness-windowed external data, limit reservation coordination, session state.
Object storage	S3-compatible, in-Kingdom, with object lock	Documents and evidence require write-once retention.
Policy execution	Embedded rules engine with versioned, effective-dated policy documents	Policy must be authored and approved outside the release cycle and remain replayable for audit.
Workflow	Durable workflow engine for long-running sagas	Sequencing spans hours or days across external systems; this must survive restarts without bespoke state management.
Frontend	React with TypeScript; Next.js for server-rendered surfaces	Mature right-to-left ecosystem, strong accessibility tooling, and a large local talent pool.
Document platform	Nutrient Document Engine, self-hosted	Generation, viewing, OCR, redaction, comparison and PAdES signing in one self-hostable component — the only way to satisfy residency without assembling four vendors.
Core banking	Tuum, behind an adapter	Party, accounts, ledger and payments. Replaceable by construction.

5. Data Design

5.1 Data principles

Ref	Principle	Consequence
DP-01	No interest construct exists	There is no rate, accrual, compounding or interest column anywhere in the schema. Return is a profit <i>amount</i> , stored once. A developer cannot accidentally compute <i>riba</i> because the field to store it does not exist.
DP-02	Goods are first-class	The model represents goods, ownership, possession, title and custody. This is a trading system that produces a financing outcome, not a lending system with goods attached.
DP-03	Evidence is append-only	Evidence records are never updated and never deleted. Corrections are new records that supersede, with both retained.
DP-04	Immutability after execution	An executed leg is immutable. Its total is immutable. Enforced by database constraint and trigger, not by application discipline alone.
DP-05	Tenant on every row	Every table carries a tenant identifier with row-level security. No query can cross a tenant boundary, including by developer error.
DP-06	Bilingual by structure	Any attribute appearing in a contract or a user interface has an Arabic and an English form modelled explicitly, with Arabic designated governing for contractual content.
DP-07	Effective-dated configuration	Policies, templates, registers and parameters are versioned with effective dates. Historic decisions replay against the version in force at the time.
DP-08	Classified at rest	Every attribute carries a classification that drives encryption, masking, retention, residency and export behaviour.

5.2 Conceptual domain model

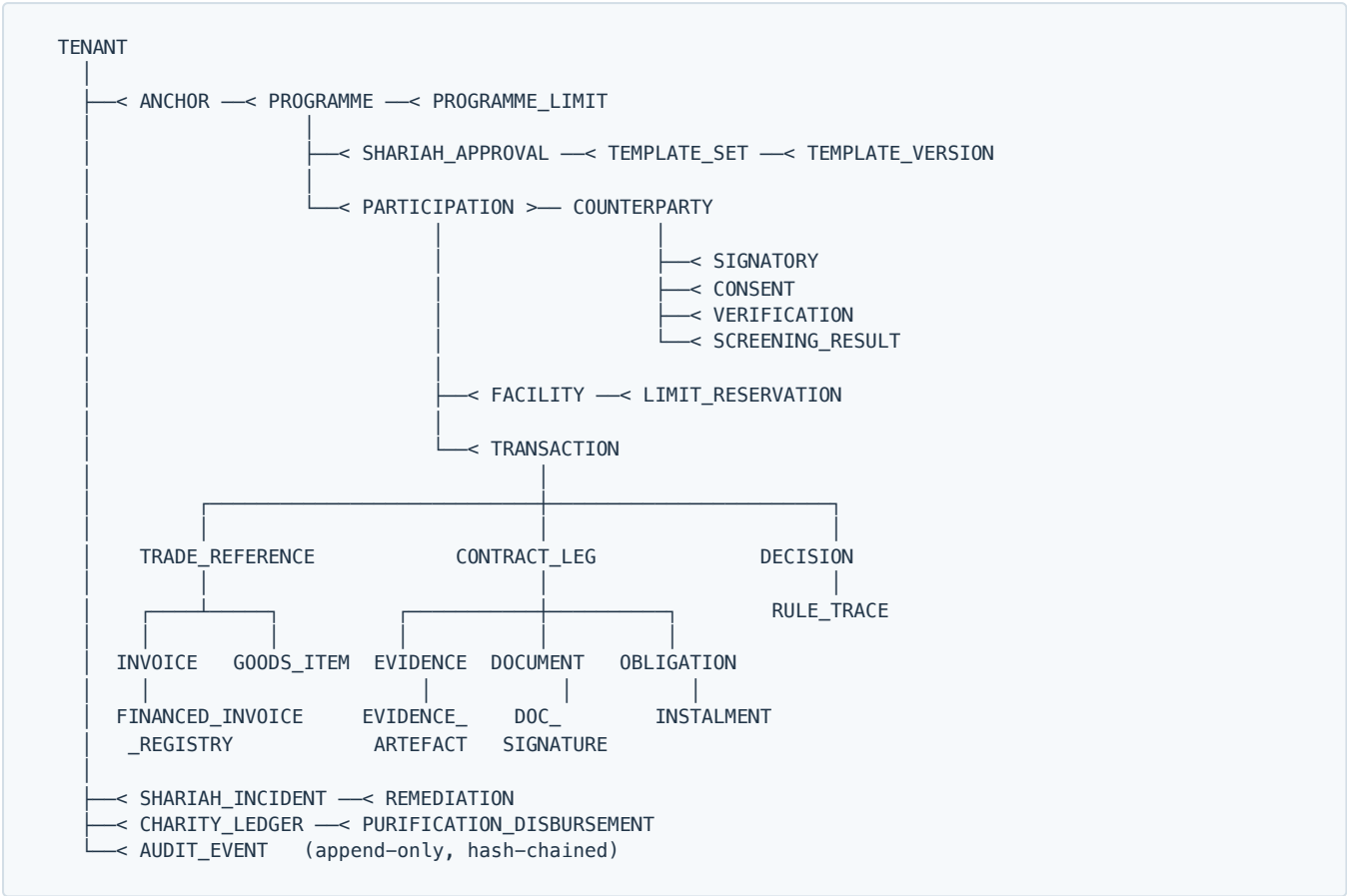


Figure 5.1 — Conceptual model. Note the absence of any rate or accrual entity.

5.3 Logical data model — entity summary

Entity	Volume profile	Purpose
tenant	Units	Deploying institution. Root of every scope.
anchor	Tens	Large corporate at the centre of a supply chain.
programme	Tens	Commercial container: product, structure, limits, pricing basis, registers, recourse.
shariah_approval	Tens	Fatwa record binding a product and template set. Gates programme activation.
template_set / template_version	Hundreds	Document templates with locked and variable regions, bound to an approval.
counterparty	Thousands	SME legal entity.
signatory	Thousands	Authorised individuals with verified authority.
consent	Tens of thousands	Granular, withdrawable PDPL consent per purpose.
verification	Tens of thousands	Registry, identity and workforce verification results with freshness.
screening_result	Tens of thousands	Sanctions, PEP, adverse media, activity permissibility.

Entity	Volume profile	Purpose
facility	Thousands	Counterparty limit within a programme.
limit_reservation	Millions	Short-lived hold during drawdown. Correctness-critical under concurrency.
transaction	Millions	The drawdown aggregate. The hot path.
trade_reference	Millions	The real trade: purchase order or cleared invoice.
goods_item	Tens of millions	Line-level goods with classification and permissibility status.
financed_invoice_registry	Millions	Permanent record of financed invoice identifiers. Never purged.
contract_leg	Millions	One leg of a structure. Immutable once executed.
evidence / evidence_artefact	Tens of millions	Typed proof of ownership, possession, delivery, identity. Append-only.
document / document_signature	Tens of millions	Generated instruments and their execution records.
obligation / instalment	Millions	The fixed amount owed and its schedule.
decision / rule_trace	Millions	Reproducible decision records with full trace.
shariah_incident	Hundreds	Compliance breaches with remediation.
charity_ledger	Thousands	Segregated non-income amounts and their disbursement.
audit_event	Hundreds of millions	Append-only, hash-chained record of everything.

5.4 Core entity specifications

5.4.1 transaction

Attribute	Type	Notes
transaction_id	UUID	Primary key. Surfaced in the user interface for support traceability.
tenant_id, programme_id, counterparty_id	UUID	Scope. Row-level security applies on tenant.
structure_code	Enum	MURABAHA, SALAM, ISTISNA, IJARAH, MUSHARAKA. Determines the sequence definition applied.
state	Enum	See 5.5.1. The compliance gate lives here.
cost_amount	Decimal(18,2)	What the institution paid for the goods.
profit_amount	Decimal(18,2)	Fixed at inception. Immutable after execution, by trigger.
sale_price_amount	Decimal(18,2)	cost + profit. The total owed. Never increases.
currency	Char(3)	SAR at launch.
tenor_days	Integer	Determinate at inception.

Attribute	Type	Notes
maturity_date_g / maturity_date_h	Date / Hijri	Both calendars stored, not derived at read time.
risk_period_start_at	Timestamp (TSA)	Possession confirmation, from a trusted time source.
risk_period_required_s	Integer	Snapshot of the Board-set minimum in force at inception.
sale_offered_at	Timestamp (TSA)	Constraint: must exceed risk_period_start_at + risk_period_required_s .
shariah_approval_id	UUID	The approval in force at execution. Snapshotted, not referenced live.
policy_version	String	Decision policy version applied.
correlation_id	UUID	Threads every log, trace, event and document.

EXPLICITLY ABSENT FROM THIS TABLE

interest_rate , profit_rate , apr , accrued_interest , compounding_frequency , penalty_rate , rate_index , margin . A benchmark may inform how profit_amount is *calculated at quotation time*, but no rate is ever persisted against an executed transaction, and nothing recomputes an amount after execution. The absence is the control.

5.4.2 contract_leg

Attribute	Type	Notes
leg_id , transaction_id	UUID	Identity and parent.
leg_type	Enum	WAAD , PURCHASE , SALE_OFFER , ACCEPTANCE , DELIVERY , TITLE_TRANSFER .
sequence_no	Integer	Position in the structure's declared sequence. Unique per transaction.
document_id	UUID	Exactly one document per leg. A document may reference only one leg — the constraint that makes combining contracts impossible.
content_hash	SHA-256	Hash of the executed document rendition.
prev_leg_hash	SHA-256	Hash chain to the preceding leg. Verifiable independently of application logs.
executed_at	Timestamp (TSA)	From a trusted timestamping authority. Constraint: strictly greater than the preceding leg's value.
tsa_token	Bytea	The RFC 3161 token itself, retained for independent verification.
template_version_id	UUID	Exactly which approved template produced this instrument.
counterparty_role , counterparty_cr	Enum / String	Role and verified registration identifier. Used for distinctness enforcement.
immutable	Boolean	Set true on execution. A database trigger rejects any subsequent update.

5.4.3 evidence

Attribute	Type	Notes
evidence_id , transaction_id , leg_id	UUID	Identity and linkage.
evidence_type	Enum	OWNERSHIP_INVOICE , TITLE_RECORD , DELIVERY_NOTE , WAREHOUSE_RECEIPT , CONSTRUCTIVE_POSSESSION , IDENTITY_ASSERTION , SETTLEMENT_CONFIRMATION . Each type has declared validation.
gate_satisfied	Enum	Which sequencing gate this evidence discharges.
source	Enum	ZATCA , REGISTRY , ANCHOR_SYSTEM , UPLOAD , OCR , PARTNER . Third- party sources rank above self-attestation.
artefact_uri , artefact_hash	String / SHA-256	Object store reference under write-once retention, with integrity hash.
captured_at	Timestamp (TSA)	When the fact became true, not when it was recorded.
validation_status , validation_detail	Enum / JSONB	Outcome of type-specific validation.
superseded_by	UUID, nullable	Corrections supersede. Nothing is ever deleted or overwritten.

5.4.4 financed_invoice_registry

Small, simple and among the most important tables in the system. It carries the tenant, the invoice unique identifier from the e-invoicing authority, the invoice hash, the issuer and recipient registration numbers, the financed amount, the transaction identifier and the financed timestamp. A unique constraint on (tenant_id, invoice_uuid) makes duplicate financing a database-level impossibility rather than an application check. Records are never purged, including after settlement, and including beyond the retention period of the transaction itself.

5.4.5 charity_ledger

Segregates every amount that must not be recognised as income: late payment charges, non-compliant income identified by review, and incidental impermissible receipts. Carries the source transaction, the amount, the reason, the computation basis, the Board-approved recipient, the disbursement reference and date, and the approval record. **The chart of accounts mapping for this ledger has no path to a revenue account.** This is enforced in the accounting configuration, not in application logic, so that no code change can route these amounts to income.

5.5 State machines

5.5.1 Transaction state machine — Murabaha

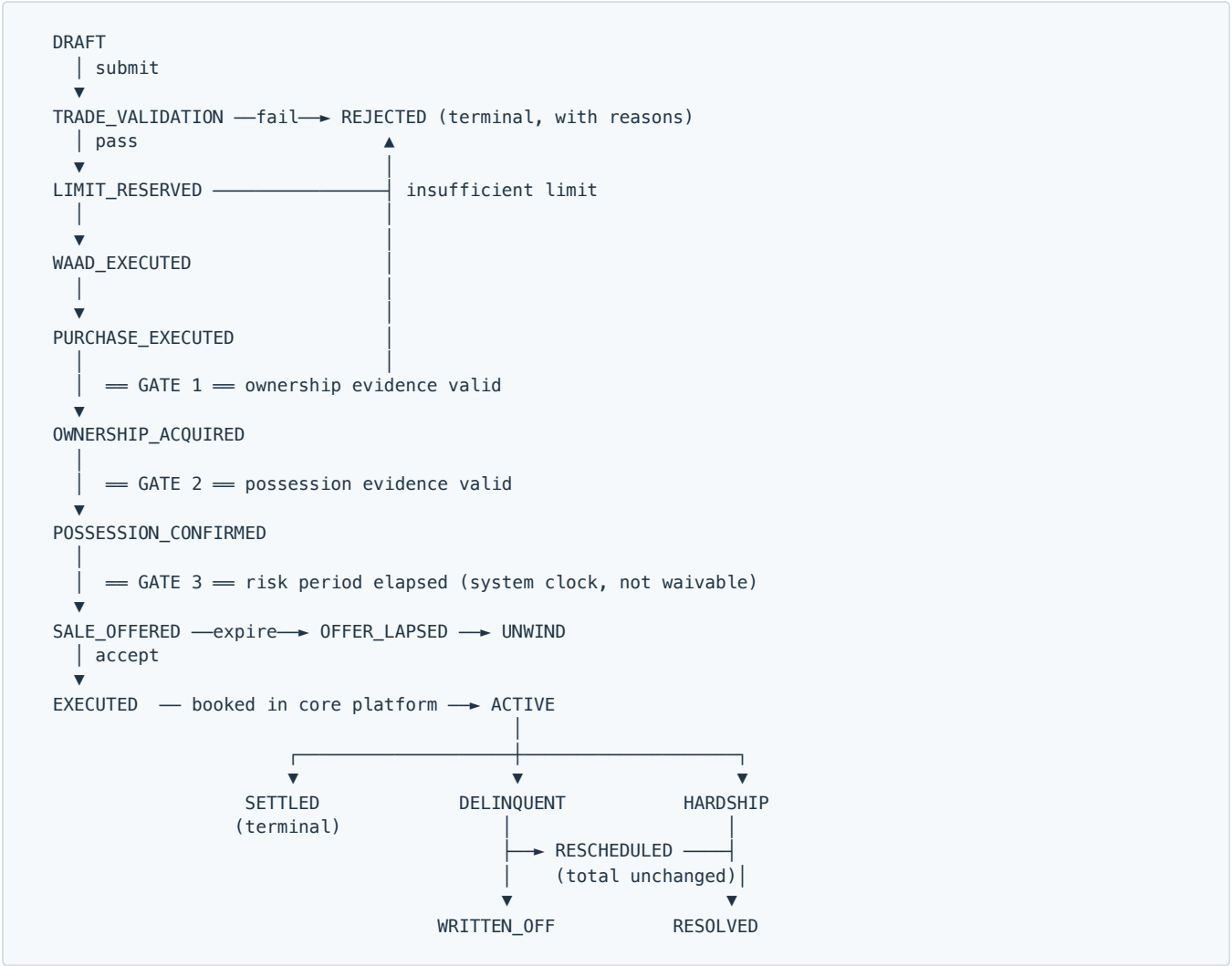


Figure 5.2 — Transaction lifecycle. No transition exists from `PURCHASE_EXECUTED` or earlier directly to `SALE_OFFERED`. This is the structural enforcement of SH-05 and SH-06.

5.5.2 Other state machines

Entity	States
counterparty	INVITED → VERIFYING → SCREENING → DECISIONING → PENDING_AGREEMENT → ACTIVE , with branches to MANUAL_REVIEW , SUSPENDED , DECLINED , OFFBOARDED
programme	DRAFT → PENDING_SHARIAH_APPROVAL → APPROVED → ACTIVE → SUSPENDED → CLOSED . Approval lapse forces SUSPENDED automatically.
document	QUEUED → GENERATED → PRESENTED → EXECUTED → SEALED → ARCHIVED , with DRIFT_DETECTED raising a Shariah incident.
shariah_incident	RAISED → TRIAGED → REMEDIATION_PLANNED → REMEDIATED → CLOSED , with a mandatory purification branch where income was affected.
limit_reservation	HELD → CONSUMED or RELEASED or EXPIRED . Time-bounded; expiry is automatic.

5.6 Evidence and immutability design

Mechanism	Implementation
Hash chaining	Each executed leg stores its own content hash and its predecessor's. The chain is anchored to the master agreement. Verification requires only the documents themselves, independent of the database.
Trusted timestamps	Every leg carries an RFC 3161 token from an external timestamping authority, retained in full. Sequence order is provable against a third party's clock, not the institution's.
Write-once storage	Documents and evidence artefacts are written to object storage under an object-lock retention policy matching the statutory period. Deletion is prevented by the storage layer.
Database immutability	Executed legs, obligations and evidence carry an immutability flag. Update and delete triggers reject any change. Corrections create superseding records.
Audit chain	<code>audit_event</code> is append-only and hash-chained, with periodic checkpoint anchoring. Tampering is detectable without trusting the application.
Replayability	Decisions replay from the persisted input snapshot and policy version. Transactions replay from the event log. Neither depends on current configuration.

5.7 Reference and configuration data

Configuration	Owner	Versioning
Structure sequence definitions	Shariah Board	Versioned, effective-dated, bound to an approval record
Minimum risk-holding periods	Shariah Board	Per product; snapshotted onto each transaction
Excluded-goods register	Shariah Board	Effective-dated; line-item classification codes
Excluded-activity register	Shariah Board	Effective-dated; activity classification codes
Approved charity recipients	Shariah Board	Effective-dated
Early settlement relief basis	Shariah Board + Compliance	Effective-dated formulation, never code
Credit policy and scorecards	Credit Risk	Versioned; champion and challenger; replayable
Concentration limits	Credit Risk	Effective-dated by dimension
Document templates	Legal + Shariah Board	Versioned with locked and variable regions; bound to an approval
Programme parameters	Business	Versioned per programme
Screening lists	Compliance	Provider-maintained, refresh tracked
Entitlements and roles	Security	Versioned; no gate-bypass entitlement may be defined

Board ownership and multi-board scope. Every row above owned by the Shariah Board is tenant-scoped and bound to the approval record of *that institution's* Board. Where the two institutions' Boards rule differently (Section 3.11), the difference is expressed entirely in this configuration layer — no code path distinguishes one institution from the other. Each transaction snapshots the parameter values in force at execution, so a Board auditing a historic transaction sees the ruling it was executed under, not the current one.

5.8 Data classification and PDPL

Class	Examples	Handling
Restricted	National identifiers, identity assertions, signature key material, biometric references	Column-level encryption, HSM-held keys, masked by default in every interface, access logged individually, never in logs or non-production
Confidential	Financial data, credit assessments, decisions, contract terms, pricing	Encrypted at rest, role-scoped access, redacted before third-party disclosure
Internal	Programme configuration, aggregate metrics, operational data	Standard controls
Public	Product descriptions, published rates and terms	No restriction

PDPL obligations in the data model

- **Lawful basis per purpose** recorded on the `consent` entity, granular and independently withdrawable. Withdrawal is honoured immediately and propagates to every dependent process.
- **Data subject rights.** Access, correction and erasure supported by design. Erasure is constrained by statutory retention: the system distinguishes data that must be retained by law from data that must be erased on request, and documents the basis for each. The `financed_invoice_registry` retains only the minimum identifiers necessary to prevent duplicate financing, which is a legitimate and documented retention basis.
- **Purpose limitation** enforced technically: data acquired under one consent scope is not readable by a process operating under another.
- **Cross-border transfer:** none. No processor, log sink, analytics service or backup destination outside the Kingdom.
- **Breach notification** workflow with defined detection, assessment, notification and record-keeping steps.

5.9 Retention, archival and residency

Data	Retention	Form
Executed contracts and evidence	Statutory period from final settlement — to be confirmed with the client's legal function; assume ten years for planning	Long-term preservation PDF with embedded validation material, under object lock
Decision records and rule traces	Same as the related transaction	Structured, replayable
Audit events	Same as the related transaction, minimum	Append-only, hash-chained

Data	Retention	Form
Financed invoice registry	Indefinite	Minimised identifiers only
Screening results	Per AML record-keeping requirements	Structured with provider reference
Declined applications	Shorter period, then minimised	Reason codes retained, personal data minimised
Operational logs	Short, per the retention schedule	Restricted data redacted at source

Residency. All primary storage, replicas, backups, object storage, log aggregation, analytical stores and third-party processors are in-Kingdom. This constrains managed service selection and is verified as a continuous control, not a one-time design assertion. The document platform is self-hosted inside the same trust boundary for exactly this reason.

5.10 Analytical data

Operational stores are not queried for analysis. Events from the backbone populate an in-Kingdom analytical store with conformed dimensions for tenant, anchor, programme, counterparty, goods category, structure and time. Restricted attributes are pseudonymised on the way in. The analytical layer serves four distinct audiences: portfolio and risk reporting, anchor programme reporting, Shariah compliance reporting to the Board, and model development for credit policy. Access is scoped separately for each; the Board's compliance dashboards do not expose commercial pricing, and the commercial dashboards do not expose personal data.

6. Backend Design

6.1 Technology stack

Layer	Selection	Note
Runtime	Kotlin on the JVM with Spring Boot (or .NET, aligned to client capability)	Mature cryptography, HSM and PDF interoperability; strong null-safety and sealed types, which the domain model relies on to make illegal states unrepresentable.
Persistence	PostgreSQL with row-level security; JSONB for evidence payloads and policy documents	Transactional guarantees carry the sequencing invariants.
Migrations	Versioned, forward-only, reviewed	Immutability triggers are part of the schema, not application code.
Messaging	Kafka with schema registry; Avro or Protobuf contracts	Ordered per transaction partition; replayable.
Workflow	Durable workflow engine (Temporal or equivalent)	Sequencing spans hours to days across external systems and must survive restarts.
Cache	Redis	Freshness windows, limit reservation coordination, idempotency keys.
Object storage	S3-compatible with object lock	Write-once retention for documents and evidence.
API	REST with OpenAPI 3.1, generated clients, contract tests	GraphQL is explicitly not used at the edge; a regulated audit surface benefits from explicit, enumerable endpoints.
Policy	Embedded rules engine over versioned policy documents	Authored and approved outside the release cycle.

6.2 Service catalogue and responsibilities

Service	Owns (data)	Key operations
programme-service	anchor, programme, programme_limit	Create and activate programmes; guard activation on Shariah approval; manage invitations.
counterparty-service	counterparty, signatory, consent, verification	Onboarding orchestration; signatory authority verification; consent lifecycle; periodic re-verification.
screening-service	screening_result, registers	Sanctions, PEP, adverse media; business activity and goods permissibility; scheduled rescreening.
decision-service	decision, rule_trace	Evaluate policy; produce a reproducible decision with reason codes; champion-challenger.
limit-service	facility, limit_reservation	Assign, reserve, consume, release; concentration enforcement.

Service	Owns (data)	Key operations
transaction-service	transaction, trade_reference, goods_item, financed_invoice_registry	Initiate drawdown; validate trade; register financed invoices; three-way match.
sequencing-service	contract_leg, structure definitions	Advance the state machine; evaluate gates; enforce risk period; chain hashes.
evidence-service	evidence, evidence_artefact	Capture, validate and verify typed evidence; integrity checking.
document-service	document, document_signature, template bindings	Generate, present, seal, archive, compare against template.
settlement-service	settlement instructions, reconciliation	Instruct disbursement and collection; reconcile; handle returns.
shariah-service	shariah_approval, shariah_incident, charity_ledger	Approval lifecycle; continuous review; incidents; purification; audit workspace queries.
audit-service	audit_event	Append, chain, verify, export.
notification-service	notification, delivery records	Arabic-first templated messaging with delivery evidence.

6.3 API design standards

Ref	Standard
BE-01	REST over HTTPS, resource-oriented, OpenAPI 3.1 as the contract of record. The specification is authored first and the implementation verified against it by contract test.
BE-02	Versioned by URI major version. Breaking changes require a new major version; additive changes do not.
BE-03	Every state-changing request accepts an Idempotency-Key header. Replays return the original result, never a duplicate effect.
BE-04	Every request and response carries a correlation identifier, propagated to logs, traces, events and generated documents.
BE-05	Errors follow RFC 9457 problem details, with a machine-readable code and a bilingual human-readable message. Compliance rejections state the specific control that blocked the request.
BE-06	Monetary values are transmitted as minor-unit integers with an explicit currency. No floating point anywhere in the financial path.
BE-07	Dates carry both Gregorian and Hijri representations where they appear in contracts or user interfaces.
BE-08	Collections are cursor-paginated. Offset pagination is not used on high-volume resources.
BE-09	Tenant scope is derived from the authenticated principal and never accepted from the client.
BE-10	Rate limits applied per principal and per tenant, with clear headers and predictable retry semantics.

6.4 Representative API specifications

6.4.1 Initiate a drawdown

```
POST /v1/transactions
Idempotency-Key: 8f2c1a4e-...
Authorization: Bearer ...

{
  "programmeId": "prg_01H...",
  "counterpartyId": "cpt_01H...",
  "structureCode": "MURABAHA",
  "tradeReference": {
    "type": "ZATCA_INVOICE",
    "invoiceUuid": "3cf5d9a2-...",
    "invoiceHash": "NWZLY2ViNjZmZj...",
    "issuerCr": "1010XXXXXX",
    "recipientCr": "4030XXXXXX"
  },
  "requestedAmount": { "value": 18500000, "currency": "SAR" },
  "requestedTenorDays": 90
}

201 Created
{
  "transactionId": "txn_01H...",
  "state": "TRADE_VALIDATION",
  "correlationId": "cor_01H...",
  "quote": {
    "costAmount": { "value": 18500000, "currency": "SAR" },
    "profitAmount": { "value": 462500, "currency": "SAR" },
    "salePriceAmount": { "value": 18962500, "currency": "SAR" },
    "maturityDateGregorian": "2027-01-04",
    "maturityDateHijri": "1448-07-25",
    "disclosure": {
      "ar": "التكلفة ١٨٥ ٠٠٠ ريال، الربح ٤٦٢ ٠٠٠ ريال، الإجمالي ١٨٩٦٢ ٠٠٠ ريال",
      "en": "Cost SAR 185,000. Profit SAR 4,625. Total SAR 189,625."
    }
  },
  "_links": { "self": "/v1/transactions/txn_01H..." }
}
```

The quote carries a profit *amount* and a total, never a rate. The disclosure block is generated from the same values that will be rendered into the contract, so the screen and the instrument cannot disagree.

6.4.2 A compliance rejection

```
409 Conflict
Content-Type: application/problem+json

{
  "type": "https://sanad.io/problems/duplicate-invoice-financing",
  "title": "Invoice already financed",
  "status": 409,
  "code": "SH-10_DUPLICATE_FINANCING",
  "detail": {
    "ar": "تم تمويل هذه الفاتورة مسبقاً ولا يمكن تمويلها مرة أخرى",
    "en": "This invoice has already been financed and cannot be financed again."
  },
  "control": "SH-10",
  "existingReference": "txn_01G...",
  "correlationId": "cor_01H..."
}
```

Every compliance rejection names the control that produced it. This matters for audit and it matters for support: the reason is never a generic decline.

6.4.3 Record ownership evidence

```
POST /v1/transactions/{transactionId}/evidence
Idempotency-Key: ...

{
  "evidenceType": "OWNERSHIP_INVOICE",
  "gate": "GATE_1_OWNERSHIP",
  "source": "ZATCA",
  "payload": {
    "invoiceUuid": "7ab1...",
    "recipientCr": "7001XXXXXX",    // the institution
    "issuerCr": "1010XXXXXX",      // the anchor
    "clearanceStatus": "CLEARED"
  },
  "artefact": { "uploadId": "upl_01H..." }
}

202 Accepted
{
  "evidenceId": "evd_01H...",
  "validationStatus": "VALID",
  "gateStatus": {
    "GATE_1_OWNERSHIP": "SATISFIED",
    "GATE_2_POSSESSION": "PENDING",
    "GATE_3_RISK_PERIOD": "NOT_STARTED"
  },
  "transactionState": "OWNERSHIP_ACQUIRED"
}
```

6.4.4 Attempting to bypass a gate

```
POST /v1/transactions/{transactionId}/legs/sale-offer

422 Unprocessable Entity
{
  "type": "https://sanad.io/problems/sequence-violation",
  "title": "Sale leg not reachable from current state",
  "status": 422,
  "code": "SH-05_SEQUENCE_VIOLATION",
  "detail": {
    "ar": "لا يمكن عرض البيع قبل تملك البضاعة وقبضها وانقضاء مدة الضمان",
    "en": "The sale cannot be offered before ownership, possession and the risk-holding period are complete."
  },
  "currentState": "PURCHASE_EXECUTED",
  "unsatisfied": ["GATE_1_OWNERSHIP", "GATE_2_POSSESSION", "GATE_3_RISK_PERIOD"]
}
```

NOTE ON THIS RESPONSE

This is not a permission error and there is no elevated role that turns it into a success. The transition does not exist in the state machine from this state. The endpoint returns 422 because the request is semantically impossible, not because the caller is insufficiently privileged.

6.5 Sequencing Engine implementation

6.5.1 Declarative structure definitions

Structures are data, not code. A definition names the legs, their order, the gates between them, the evidence types that discharge each gate, and the minimum risk period. It is bound to a Shariah approval record and is versioned. Adding a structure is a configuration and approval activity; it does not require an engine change.

```
structure: MURABAHA_DISTRIBUTOR
version: 3
shariahApprovalRef: SSB-2026-014
legs:
  - type: WAAD          seq: 1  document: waad_promise
  - type: PURCHASE      seq: 2  document: purchase_contract
  - type: SALE_OFFER    seq: 3  document: murabaha_offer
  - type: ACCEPTANCE    seq: 4  document: murabaha_acceptance
gates:
  - id: GATE_1_OWNERSHIP
    after: PURCHASE
    requires:
      anyOf: [OWNERSHIP_INVOICE, TITLE_RECORD]
      validation: recipientCr == institution.cr AND clearanceStatus == CLEARED
  - id: GATE_2_POSSESSION
    after: GATE_1_OWNERSHIP
    requires:
      anyOf: [DELIVERY_NOTE, WAREHOUSE_RECEIPT, CONSTRUCTIVE_POSSESSION]
  - id: GATE_3_RISK_PERIOD
    after: GATE_2_POSSESSION
    minimumSeconds: 3600          # Board-set; snapshotted per transaction
    clock: TRUSTED_TIMESTAMP_AUTHORITY
constraints:
  - distinctParties: [PURCHASE.seller, ACCEPTANCE.buyer] # no 'inah
  - oneDocumentPerLeg: true                               # no combined contracts
  - monotonicTimestamps: true
```

6.5.2 Implementation rules

- **Illegal states are unrepresentable.** The transaction aggregate is modelled with sealed types per state, each exposing only the transitions legal from it. A sale-offer transition does not exist as a callable operation on a `PurchaseExecuted` state. This is a compile-time guarantee, not a runtime check.
- **Gate evaluation is pure.** Given a transaction and its evidence set, gate evaluation is a deterministic function with no side effects, which makes it exhaustively testable and independently re-runnable by the Shariah review process.
- **The clock is external.** Risk period elapse is evaluated against the timestamping authority's time, never the application server's, and never a client-supplied value.
- **Chaining is synchronous with execution.** A leg is not marked executed until its document hash, its predecessor hash and its timestamp token are all persisted in the same transaction.
- **Durable orchestration.** The sequence runs as a durable workflow so that a process restart during a multi-hour risk period, or during an external settlement, resumes correctly rather than stranding the transaction.

6.6 Decision Engine

The engine evaluates a versioned policy document against an immutable input snapshot and returns a decision, a limit, a score, a rule trace and reason codes. Four properties are non-negotiable:

1. **Determinism.** Same snapshot plus same policy version yields the same outcome, always. No wall-clock reads, no random sampling, no live external calls during evaluation — all external data is gathered into the snapshot

beforehand.

2. **Full trace.** Every rule evaluated, its inputs, and its result are persisted. This is the audit defence and the source of customer-facing reasons.
3. **Bilingual reason codes.** Decline and limit-reduction reasons are derived from the trace and rendered in Arabic from a maintained catalogue, never composed ad hoc.
4. **Champion–challenger.** A challenger policy evaluates the same snapshot in parallel, its result recorded but not applied, enabling policy change to be measured before it is adopted.

Primary inputs at Phase 1: verified trade history from the e-invoicing authority (volume, frequency, counterparty concentration, dispute and credit-note patterns), programme parameters and anchor recourse, commercial registration age and standing, credit bureau obligations and defaults, workforce signals, and — where consented — open banking cash flow.

6.7 Limit and facility engine

The highest-concurrency component in the system, and the one where a correctness failure is most expensive. Design points:

- **Reservation before work.** A drawdown reserves limit before any external side effect. Reservations are time-bounded and expire automatically, so an abandoned journey cannot strand capacity.
- **Serialisable reservation.** Reservation against a facility is performed under a serialisable isolation guarantee or an equivalent optimistic-locking retry. Two concurrent drawdowns cannot both consume the last available capacity.
- **Hierarchical checks.** Counterparty facility, then programme limit, then concentration limits by anchor, sector and goods category — all evaluated in one atomic reservation.
- **Exposure aggregation.** Where a counterparty participates in multiple programmes, exposure aggregates across them, and where the core banking platform holds other exposure to the same entity, that is pulled in through the adapter.
- **Recourse-aware.** Where an anchor provides buyback or recourse, the effective exposure calculation reflects it per programme configuration.

DEPENDENT ON OI-02

Whether this engine lives in Sanad or is delegated to the core banking platform depends on that platform's revolving-facility capability under supply chain finance volume and concurrency. The design assumes Sanad owns it, which is the safe assumption; if the core proves capable, the engine becomes a thin adapter over it.

6.8 Document Service and Nutrient integration

Function	Implementation
Template binding	Sanad resolves the approved template version in force for the programme and structure, and snapshots the identifier onto the leg. Rendering is always against an explicit version, never "latest".
Generation	Asynchronous job. Bilingual render with Arabic as governing text, embedded fonts, full shaping and bidirectional layout, Hijri and Gregorian dates, and the approval reference in both the visible footer and the document metadata.
Locked regions	Board-approved clauses are non-substitutable regions in the template. The rendering path has no mechanism to alter them; only declared merge fields are substitutable.

Function	Implementation
One leg, one document	The service rejects any request to render more than one leg into a single instrument. This is a guard rail at the service boundary, not a convention.
Presentation	Embedded viewer in the counterparty surface with scroll-completion tracking and per-clause acknowledgement where required. The rendition presented is the rendition hashed and retained.
Signature	Master agreements: full PAdES with long-term validation via the certification provider's signing interface. Per-transaction legs: electronic acceptance bound to the identity assertion, sealed with the institutional certificate held in the HSM.
Timestamping	Every executed instrument receives an RFC 3161 token, retained in full alongside the document.
Drift comparison	A scheduled job compares executed documents against their approved template version and raises a Shariah incident on deviation. At this volume it runs unattended, continuously.
OCR	Applied to uploaded unstructured evidence, with confidence scoring; low confidence routes to an operations queue rather than silently accepting an extraction.
Redaction	Rule-driven per recipient class — guarantor, takaful operator, external counsel — applied automatically before any external disclosure.
Archival	Long-term preservation format with validation material embedded, written to object storage under retention lock.

PHASE 0 SPIKES — OI-05, OI-06

Two spikes must complete before design freeze. First, **Arabic rendering fidelity**: render the most complex real template — mixed Arabic and English, tables, both calendars side by side, Arabic-Indic and Latin numerals — and have Legal read the output. Arabic is the governing text; this is a correctness requirement, not typography. Second, **a validated signed document**: obtain a test certificate from the selected certification provider and produce one long-term-validation PDF that verifies. Until that file exists, the signature design is a hypothesis.

6.9 Integration adapters

Every adapter implements a capability-named interface and exposes no vendor vocabulary upward. Each ships with a recorded fixture set for deterministic testing, a circuit breaker, a declared freshness window, a declared failure posture (per Section 4.9), and a contract test run against the vendor sandbox in the pipeline.

```
interface CoreBankingProvider {
    resolveOrCreateParty(party: PartyDetails): PartyRef
    resolveAccount(partyRef: PartyRef, purpose: AccountPurpose): AccountRef
    bookObligation(obligation: Obligation, idempotencyKey: Key): BookingRef
    instructSettlement(instruction: SettlementInstruction, idempotencyKey: Key): SettlementRef
    fetchExposure(partyRef: PartyRef): Exposure
    subscribeLifecycle(handler: LifecycleHandler)
}

interface EInvoicingProvider {
    // ZATCA today; portable by design
    validateInvoice(ref: InvoiceRef): InvoiceValidation
    fetchTradeHistory(cr: RegistrationNumber, window: Period): TradeHistory
}
```

The `Obligation` passed to `bookObligation` carries a fixed total and a schedule. It carries no rate. If a core banking platform's API requires a rate field, the adapter supplies a zero or a structural equivalent and the fact is recorded as a known deviation — but the concept never propagates into the Sanad domain model.

6.10 Consistency patterns

Pattern	Application
Transactional outbox	Every external side effect — core banking booking, settlement instruction, event publication — is written to an outbox in the same database transaction as the state change, then dispatched by a relay. Nothing is lost if a process dies between commit and call.
Idempotency keys	Persisted with the response, so a replay returns the original outcome. Mandatory on every external write. A double-booked Murabaha is a regulatory incident, not a defect.
Durable saga	The sequencing workflow orchestrates multi-system, multi-hour processes with explicit compensation for each step. Unwind of an acquired-but-unsold position is a first-class compensating path with documented goods disposition.
Reconciliation	Nightly: every Sanad transaction in an executed state has exactly one core banking obligation, and vice versa; every settlement instruction has a confirmed or explicitly failed outcome; every financed invoice appears exactly once in the registry. Drift alerts rather than self-heals.
Event idempotency	All consumers are idempotent and tolerate at-least-once delivery and out-of-order arrival within the guarantees of their partition.

6.11 Scheduled and batch processing

Job	Cadence	Purpose
Core banking reconciliation	Nightly	Exactly-once verification in both directions
Continuous Shariah review	Continuous + nightly sweep	Sequence integrity, evidence completeness, risk-period compliance, distinctness
Template drift comparison	Continuous	Executed documents against approved template versions
Limit recalculation	Daily + on trigger	Refresh counterparty limits on new data
Rescreening	Per compliance schedule	Sanctions, PEP, adverse media refresh
Registry re-verification	Per freshness window	Commercial registration and signatory authority currency
Purification computation	Monthly	Identify, compute and segregate non-compliant amounts
Reservation expiry	Continuous	Release abandoned limit holds
Archival	Daily	Long-term preservation conversion and retention locking
Retention enforcement	Monthly	Minimise and purge per the retention schedule

6.12 Backend security implementation

- **Zero trust between services.** Mutual TLS with workload identity. No service is trusted because of its network position.
- **Central authorisation.** Entitlements are evaluated by a single policy component and enforced at each service boundary. **No entitlement exists that bypasses a sequencing gate**, and a test asserts that no such entitlement can be defined.

- **Key custody.** The institutional sealing key never leaves the HSM. Application code submits a hash and receives a signature; it never handles private key material.
- **Restricted data handling.** National identifiers and financial detail are encrypted at column level, masked by default in every read path, and redacted at source before logs leave the application.
- **Non-production isolation.** Production data is never present below pre-production. Test data is synthetic, and a pipeline control blocks any restore that would violate this.
- **Supply chain.** Dependency provenance verified, images signed, software bill of materials produced per release.

6.13 Testing strategy

Layer	Focus
Unit	Domain invariants. Property-based tests assert that no sequence of legal operations can produce a transaction whose total differs from cost plus profit, or whose legs are out of order.
Adversarial compliance	A dedicated suite that <i>attempts</i> each prohibited outcome — bypass a gate, finance a duplicate invoice, execute a sale before possession, increase a total on reschedule, post a late charge to income, render two legs into one document, transact with the same entity on both sides. Each test passes only when the attempt fails. This suite is the executable form of Section 2.9 and is reviewed with the Shariah Board. The suite is parameterised per tenant configuration and runs against each institution's own Board parameters — risk period, evidence types, registers, <i>ibra'</i> basis — so that one Board's rulings are never encoded into the tests (Section 3.11).
Contract	Every adapter tested against the vendor sandbox in the pipeline, and against recorded fixtures offline.
Integration	Full sequence execution against sandboxes for the core banking platform, document platform, e-invoicing and signing.
End-to-end	Complete journeys per persona, in Arabic and English, in both text directions.
Performance	Sustained load at the Phase 1 design point with headroom verification; concurrency tests specifically targeting limit reservation.
Security	Static and dynamic analysis in the pipeline; independent penetration test before go-live and annually thereafter.
Shariah acceptance	The Board reviews live transactions in user acceptance testing, end to end, with evidence and signature validation, as a formal phase exit gate.
Accessibility	Automated and manual testing at WCAG 2.2 AA in both language directions.

7. Frontend Design

7.1 Design principles

Ref	Principle	Consequence
FP-01	Arabic-first, not Arabic-translated	Arabic is the design default. Layouts are composed right-to-left first and mirrored to left-to-right, not the other way round. Copy is written in Arabic and translated to English, so that neither reads like a translation.
FP-02	Start from the need, not the amount	The drawdown journey begins with a real trade — an order or an invoice — never with "how much do you want?". This is the most visible expression of the Shariah structure in the product, and it is a deliberate departure from conventional lending interfaces.
FP-03	Show the amount, never a rate	Cost and profit are displayed as amounts, separately and explicitly. No rate is presented anywhere, because no rate exists. This is a validity condition of the Murabaha, and it happens to be clearer for the user.
FP-04	Honest status over false progress	When a national rail is unavailable or a document is generating, the interface says so plainly in Arabic with a realistic expectation. Never a spinner without an explanation, never a technical error.
FP-05	Mobile is the primary surface for the primary user	The SME owner is on a mid-range Android phone on a constrained network. That is the design target, not a responsive afterthought.
FP-06	Declines explain themselves	Every decline, limit reduction and block carries a specific, respectful Arabic reason derived from the decision trace — never a generic refusal.
FP-07	The Board self-serves	The Shariah audit workspace is designed for scholars, not engineers. A Board member must be able to open a transaction and understand its compliance without any technical vocabulary.
FP-08	One design system, five surfaces	Tokens, components and patterns are shared. Surfaces differ in density and permission, not in language or visual identity.

7.2 Surfaces and personas

Surface	Primary persona	Form factor	Character
Wasl SME	Distributor / supplier owner or finance manager	Mobile-first web, installable	Sparse, fast, few decisions per screen, large tap targets, works on a poor connection.
Anchor Portal	Corporate treasury or commercial manager	Desktop web	Dense, data-rich, built for monitoring and reconciliation. Bulk operations and export.
Operations Workbench	Underwriter, operations and compliance officer	Desktop web	Queue-driven. Everything needed to decide one case, on one screen, with the evidence inline.
Shariah Audit Workspace	Board member, Shariah auditor and reviewer	Desktop web	Read-only by construction. Investigation-oriented. Plain language. Exportable.

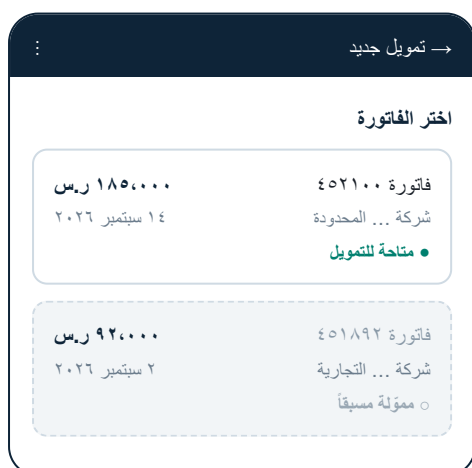
Surface	Primary persona	Form factor	Character
Admin Console	Platform administrator	Desktop web	Configuration, versioning, approval workflow, entitlements. Change is always reviewed and always audited.

7.3 Information architecture

WASL SME (mobile)	ANCHOR PORTAL	OPERATIONS WORKBENCH
- Home - Available limit - Request finance - What's due - Request - Choose the trade - Review cost + profit - Read the contract - Accept - My finance - Active - Schedule (Hijri/Greg) - Documents - Pay early - Support - Talk to us - Report hardship	- Dashboard - Programme utilisation - Counterparty status - Transactions today - Settlement position - Counterparties - Invited / Active - Onboarding progress - Invite (bulk / API) - Transactions - Live pipeline - Settled - Export - Programme - Limits & terms - Goods register - Reports	- My Queue - Onboarding exceptions - Screening hits - Evidence exceptions - Matching exceptions - Settlement failures - Case detail - Full evidence pack - Decision trace - Four-eyes action - Notes & audit - SHARIAH AUDIT WORKSPACE - Transaction search - Transaction replay - Leg timeline - Evidence per gate - Signature validation - Template & fatwa ref - Export audit pack - Incidents - Purification ledger - Approvals register

7.4 Key screens

7.4.1 SME — request finance



Step 1 — Choose the trade

The screen lists the counterparty's own cleared invoices, retrieved from the e-invoicing authority. The user selects a real trade.

Invoices already financed are shown greyed with an explicit reason rather than hidden. The user must be able to understand why something is unavailable.

The user never types an amount. The trade determines the amount. This is the Shariah structure made visible in the interface, and it is the clearest single departure from a conventional lending journey.

عرض المراجعة →

التكلفة

١٨٥,٠٠٠ ر.س

الربح

٤,٦٢٥ ر.س

الإجمالي

١٨٩,٦٢٥ ر.س

تاريخ الاستحقاق

٢٥ رجب ١٤٤٨

٤ يناير ٢٠٢٧

اقرأ العقد ومتابعة →

Step 2 — The offer

Cost and profit are shown separately and explicitly, as amounts. The total is the total, and it will never change.

No rate appears anywhere, because none exists. Disclosure of cost and markup is a validity condition of the Murabaha under SH-15; it also happens to be clearer for the user than an annualised percentage.

Both calendars are shown wherever a date has contractual effect, with the user's preference determining which leads.

The contract is then read in an embedded viewer — never downloaded, never summarised. Arabic is the governing text and is what is presented.

7.4.2 Shariah Audit Workspace — transaction replay

Transaction txn_01H8K...

Murabaha · Distributor Finance

Programme: Anchor A — Building Materials

Fatwa: SSB-2026-014 (v3)

- Wa'd — promise to purchase

14 Sep 2026 09:12:04 ✓ signed

Document · Hash 0x7f3a... · TSA verified
- Purchase from anchor

14 Sep 2026 09:14:41 ✓ sealed

Document · Hash 0x9c1b... · chained to 0x7f3a... ✓
- GATE 1 — OWNERSHIP

14 Sep 2026 09:31:08 ✓ SATISFIED

Evidence: ZATCA invoice 7ab1... · recipient = institution CR ✓

clearance CLEARED ✓ · source: ZATCA (third party)
- GATE 2 — POSSESSION

14 Sep 2026 10:02:55 ✓ SATISFIED

Evidence: Delivery note (OCR, confidence 0.97) + warehouse receipt
- GATE 3 — RISK PERIOD

required 60m · actual 91m ✓

Measured against external timestamping authority, not local clock
- Murabaha sale offer

14 Sep 2026 11:34:22 ✓ sealed

Cost 185,000 · Profit 4,625 · Total 189,625 · disclosed ✓

Template v3 · approval SSB-2026-014 · no drift ✓
- Acceptance by counterparty

14 Sep 2026 11:41:07 ✓ accepted

Identity assertion verified ✓ · chained ✓ · TSA verified ✓

Distinct parties ✓

One document per leg ✓

Timestamps monotonic ✓

Goods permissibility ✓

Invoice not previously financed ✓

[Verify full chain]

[Export audit pack]

[Raise an observation]

Figure 7.1 — The Board's view. Plain language, third-party evidence identified as such, and every Shariah control shown as satisfied or not. No technical vocabulary and no IT request required.

7.4.3 Operations workbench — case detail

Queue-driven, single-screen. The left column holds the case facts; the centre holds the exception itself with the specific control that raised it; the right column holds the full evidence pack with inline document preview. Actions are constrained by entitlement, four-eyes approval is enforced where policy requires it, and — critically — **no action available on this screen can advance a transaction past an unsatisfied sequencing gate**. Operations resolves evidence and data problems; it cannot resolve a compliance gate by deciding to.

7.4.4 Anchor portal — programme dashboard

Programme limit and utilisation, counterparty onboarding funnel, today's transaction pipeline by state, settlement position, overdue exposure, and concentration against configured limits. Bulk counterparty invitation by upload or API, and exports that reconcile to the anchor's own ledger. Dense by design: this user is monitoring, not transacting.

7.5 Arabic-first, RTL and Hijri

Concern	Approach
Direction	Logical CSS properties throughout — <code>inline-start</code> and <code>inline-end</code> , never <code>left</code> and <code>right</code> . Direction is set at the document root. Every component is developed and reviewed in both directions; a component that has not been seen in RTL is not complete.
Typography	A typeface with genuine Arabic design, not a Latin face with an Arabic fallback. Arabic and Latin optical sizes tuned independently, since matching point sizes do not produce matching perceived weight.
Bidirectional content	Arabic text containing Latin fragments — company names, invoice references, registration numbers — isolated correctly so that neighbouring punctuation and digits do not reorder. This is a recurring defect class and is covered by explicit test cases.
Numerals	Arabic-Indic or Latin digits by user preference, applied consistently within a view. Machine-readable identifiers always render in Latin digits regardless of preference, because they are copied and quoted.
Currency	Correct Saudi riyal symbol placement per direction, with thousands separation appropriate to the numeral system in use.
Calendars	Hijri and Gregorian both available, with a user preference determining which leads. Both are shown wherever a date has contractual effect. Dates are stored in both forms rather than converted at read time, so a stored contractual date can never shift.
Iconography	Directional icons mirror; non-directional icons do not. Reviewed individually rather than mirrored wholesale.
Copy	Authored in Arabic and translated to English. Legal and Shariah terminology uses the register the Board and the courts use, not a plain-language paraphrase.

7.6 Design system

Element	Definition
Tokens	Colour, type scale, spacing, radius, elevation and motion defined as tokens. Client branding is a token override, never a component fork — this is what makes the product re-skinable per client without a code change.
Theming	Light and dark, both meeting contrast requirements in both directions.
Components	A single shared library across all five surfaces, each component built and tested in both directions, with documented states including loading, empty, error, degraded and offline.
Money	A dedicated component. It renders cost, profit and total together, correctly for direction and numeral preference, and it is <i>incapable</i> of rendering a rate. There is no prop for one.
Date	A dedicated component handling dual calendars and preference.
Status	A shared vocabulary of transaction and gate states with consistent colour, icon and bilingual label, so the same state never appears differently on two surfaces.
Evidence	A shared component that displays a piece of evidence with its type, source, validation result and integrity status, used identically in operations and in the Board's workspace.

7.7 Document review and signing experience

- **Read in place.** The contract is presented in an embedded viewer within the journey. It is never a download link and never a summary — the counterparty reads the instrument they are executing.
- **Arabic is what is shown.** The governing Arabic text leads, with the English translation available alongside and labelled as a translation.
- **Completion tracking.** Scroll completion is tracked and, where the Board or Legal requires it, specific clauses require individual acknowledgement.
- **Explicit disclosure.** Cost and profit are restated in the interface immediately adjacent to the acceptance control, matching the document exactly. The values come from the same source, so they cannot diverge.
- **What is signed is what was seen.** The exact rendition presented is the rendition hashed, sealed and archived. The evidence record ties the identity assertion, the timestamp and the document hash to that specific rendition.
- **Master agreement ceremony.** Full digital signing of the master agreement is a deliberately distinct, more formal experience, reflecting that it is executed once and governs everything after it.
- **Per-transaction acceptance.** Streamlined but never silent. The counterparty always sees the instrument and always acts affirmatively. There is no one-click path that skips presentation.

7.8 Accessibility

WCAG 2.2 level AA on every surface, verified in both language directions. Full keyboard operability with a visible focus order that follows reading direction. Semantic structure and correct language attributes so that Arabic screen readers announce correctly — including where Latin fragments are embedded. Contrast verified in both themes and both directions. Touch targets sized for the mid-range device that is the SME's primary hardware. Error messages are associated programmatically with their fields and are specific rather than generic. Accessibility is tested in the pipeline and reviewed manually, not asserted.

7.9 Frontend architecture

Concern	Approach
Framework	React with TypeScript. Next.js for the server-rendered SME surface, where first-load performance on a constrained network matters most.
Repository	Monorepo: one shared design system and one shared API client, five applications. Shared code is versioned and consumed, not copied.
Backend-for-frontend	One per surface, shaping and aggregating for that surface's needs. This keeps the SME surface's payloads small and keeps surface-specific concerns out of the core services.
API client	Generated from the OpenAPI specification. Types are never hand-written, so a backend contract change breaks the build rather than production.
State	Server state managed by a query library with explicit caching and invalidation; client state kept minimal and local. Transaction state always originates from the server — the frontend never infers a gate result.
Offline and poor connectivity	The SME surface caches read data for offline viewing and queues nothing that has financial effect. A user on a weak connection can always see what they owe; they cannot accidentally submit twice.
Error handling	Compliance rejections are rendered from the control code into a specific, respectful Arabic explanation. Technical errors never reach the user.
Feature flags	Per tenant and per programme, enabling staged rollout and client-specific configuration without branching.
Analytics	In-Kingdom, consent-gated, with no restricted data in any event payload.

7.10 Performance budgets

Metric	SME surface	Internal surfaces
Largest contentful paint (mid-range Android, 4G)	< 2.5 s	< 3.0 s
Interaction to next paint	< 200 ms	< 200 ms
Initial JavaScript transferred	< 180 KB compressed	< 350 KB compressed
Arabic font payload	Subset, preloaded, < 90 KB	Subset, < 90 KB
Time to usable limit view from cold start	< 3 s	n/a

Budgets are enforced in the pipeline. A regression fails the build rather than being noticed in production by the user least able to tolerate it.

8. Delivery Plan

8.1 Phases and gates

Phase	Duration	Content	Exit gate
Phase 0 Foundation	4–6 weeks gated on Board agenda availability	Agenda slots secured with both Boards. Shariah artefact harvest and the parallel divergence exercise (3.11). Anchor commitment. Platform capability verification. Certification provider selection. Arabic rendering and signature spikes. Tax position. Licensing position. Target operating model and domain model.	Sign-off from both Boards on structures, sequence and execution pattern; divergence map complete with every divergence point named as a configuration key; signed anchor letter of intent ; written platform capability answers; one validated Arabic signed document.
Phase 1 Distributor Finance	12–16 weeks	Anchor and programme management, counterparty onboarding, decisioning and limits, sequencing engine, document and signature, settlement, core banking booking, operations workbench, Shariah audit workspace, hardship path.	Board reviews live UAT transactions end to end and issues the product fatwa; independent penetration test passed; performance at design point demonstrated.
Phase 2 Scale and structures	12–14 weeks	Supplier early payment, Salam and Istisna'a purchase order finance, multiple anchors, guarantee programme, trade credit takaful, purification automation, expanded analytics.	First internal Shariah audit passes on production data.
Phase 3 Breadth and product	14+ weeks	Inventory and Ijarah, government contract finance, import and export, growth capital, and the productised packaging for the second client.	Second client onboarded with zero core code change.

8.2 Team shape — Phase 1

Role	Count	Note
Product owner — SME trade finance	1	Must have genuine Saudi trade finance domain depth. Not substitutable.
Solution architect	1	Owns the Sanad–Tuum–Nutrient boundary and the adapter discipline.
Backend engineers	4	One dedicated to the sequencing engine throughout.
Frontend engineers	2	At least one with substantial RTL and Arabic typography experience.
Integration engineer	1	National rails full time. These are calendar-driven and relentless.
Credit policy engineer	1	Policy-as-code, scorecards, decision reproducibility.

Role	Count	Note
Document and signature specialist	1	Phase 1 heavy: Arabic rendering, PAdES, certification provider integration.
QA engineer	2	Arabic capability required. One owns the adversarial compliance suite.
DevOps / SRE	1	In-Kingdom infrastructure, SAMA control evidence.
Shariah and compliance liaison	0.5	Standing channel to the Board. Not an end-of-phase review.
Delivery manager	1	—

8.3 Immediate actions

1. **Intellectual property clause** into the client contract discussion. Before any code is written. This is the highest-value item on the list and the cheapest to secure now.
2. **Confirm the anchor.** Named corporate, letter of intent, and an explicit conversation about the procurement change that taking title requires.
3. **Request agenda slots with both Shariah Supervisory Boards.** Immediately, and ahead of finalising the question set — a holding request costs nothing and secures the calendar. Board agenda lead time is commonly four to eight weeks and cannot be compressed retrospectively. This is the most time-sensitive item on the list.
4. **Request each institution's existing Shariah artefacts** (Section 8.4). The approved contract template set is the single most valuable item and saves weeks of drafting.
5. **Six platform capability questions** to the core banking vendor, in writing, with a written answer requested — plus volume pricing.
6. **Certification provider shortlist.** Longest lead time of any dependency. Start now.
7. **Document platform licensing** for self-hosted, in-Kingdom deployment.
8. **Trademark clearance** on Sanad and Wasl at SAIP, with Mithaq and Imdad as alternates.
9. **Tax counsel** on the VAT treatment of the two supply legs.

8.4 Shariah artefact harvest

Both institutions hold sitting Boards, which means both hold existing rulings, precedents and approved documents. Phase 0 is therefore a harvest exercise before it is a ruling exercise. Request the following from each institution at the outset:

Artefact	Why it matters
Approved contract template set	The most valuable single item. It establishes the drafting register, clause structure and terminology the Board already accepts, and removes weeks of drafting and review.
Existing fatwas on Murabaha, and on Salam and Istisna'a where held	Establishes the positions already settled, so Phase 0 questions address only what is genuinely open.
Shariah governance charter and manual	Defines the approval workflow, the Board's authority and the reporting obligations the platform must satisfy.
Internal Shariah review methodology	Sanad's continuous review (2.8.6) must align with the institution's existing first- and second-line process, not replace it.
Shariah audit methodology and annual report format	Determines the output formats the audit workspace and reporting must produce.

Artefact	Why it matters
Existing positions on constructive possession, risk-holding periods, <i>ibra'</i> , late payment and purification	Direct inputs to the divergence map and to the configuration schema.
Existing excluded-goods and excluded-activity registers	Seed data for the screening registers.
Board composition, chairman, meeting calendar, agenda lead time	Determines the achievable Phase 0 exit date.

SCOPE NOTE ARISING FROM THE HARVEST

An institution with a sitting Board will already operate an internal Shariah compliance function and a Shariah audit function. The Shariah Governance module described in Section 2.8.6 must therefore **integrate with those existing functions and their reporting formats** rather than substitute for them. Obtain the report templates early; retrofitting an output format is cheap in Phase 0 and expensive in Phase 1.

9. Verification Register — Open Items

Every open item below is unresolved at the time of writing. Each is recorded here rather than assumed away, because each can change the design, the timeline or the cost. Items marked Critical sit on the path to design freeze. Closed items are retained with their resolution rather than removed, so that the register reads as a history.

Ref	Owner	Severity	Question to be answered
0I-01	Client	CLOSED	Resolved at v0.2. Both target institutions — the Islamic bank and the fintech — hold sitting Shariah Supervisory Boards. No appointment on the critical path. Residual constraint is Board agenda lead time, tracked under OI-20.
0I-02	IOTA / Tuum	Critical	Can the core banking platform hold a fixed-price deferred sale with a profit amount and no accrual; model revolving multi-drawdown facilities at supply chain volume; post late charges to a segregated charity liability; and support an AAOIFI-aligned chart of accounts? Six written questions, written answers.
0I-03	IOTA / Tuum	Critical	Commercial model at supply chain finance volumes. A per-contract price could break the unit economics. Required before Phase 1 pricing is presented.
0I-04	Shariah Board	Critical	Is electronic acceptance under a pre-executed master agreement, bound to a verified identity assertion and sealed with the institutional certificate, valid execution of a separate contract? If not, the operating model and economics change materially.
0I-05	IOTA	Critical	Arabic rendering fidelity of the document platform against the most complex real template, reviewed by Legal. Arabic is the governing text.
0I-06	IOTA	Critical	Produce one long-term-validation signed PDF using a test certificate from a licensed certification provider, and verify it. Until this file exists the signature design is a hypothesis.
0I-07	Shariah Board	High	Is constructive possession acceptable, and what evidence satisfies it? What is the minimum risk-holding period per product? Model the economics at several settings.
0I-08	Client / Tax	High	VAT treatment of the institution's purchase and sale legs, input recovery, and invoicing obligations. Materially different from a lending product.
0I-09	Industry	High	Cross-institution duplicate financing. Wasl blocks duplicates within one institution. A market-level registry does not exist. Quantify the residual risk and consider whether to propose an industry utility.
0I-10	Client / SAMA	High	Licensing position for this activity and any specific reporting obligations arising from it. Extended at v0.2: confirm whether the fintech holds its own licence or operates under a bank's. If the latter, that bank's Board may govern its products, which changes the divergence analysis in OI-19.
0I-11	IOTA	High	Technical and commercial access terms, rate limits, availability commitments and onboarding lead times for each national rail. These are calendar-driven and historically the largest schedule risk.

Ref	Owner	Severity	Question to be answered
0I-12	Shariah Board	High	Exact late payment treatment, exact early settlement relief formulation reconciled with the regulatory requirement, purification methodology, and approved charity recipients.
0I-13	Client	High	Funding model for the book — own balance sheet, committed bank line, or investor structure. It affects architecture and licensing route.
0I-14	IOTA / Client	High	Intellectual property ownership. The productisation strategy is void if the client owns the core.
0I-15	Shariah Board	Medium	Screening thresholds for business activity permissibility, and the excluded-goods classification basis.
0I-16	Shariah Board	Medium	Treatment of the government guarantee programme, given the position on fee-bearing <i>kafala</i> .
0I-17	Client / Legal	Medium	Confirm statutory retention periods and the bilingual precedence clause wording.
0I-18	IOTA	Medium	Backend runtime selection — JVM or .NET — aligned to the client's operational capability rather than architectural preference.
0I-19	IOTA / both Boards	Critical	New at v0.2. Two-board divergence. Put the same question set to both Boards in parallel; produce the divergence map; confirm that every divergence point is absorbed by configuration and not by code. A divergence discovered during build is a redesign. See Section 3.11.4.
0I-20	Client	Critical	New at v0.2. Board agenda lead time at both institutions. Determines the achievable Phase 0 exit date and cannot be compressed retrospectively. Request slots before finalising the question set.
0I-21	Client	High	New at v0.2. Delivery of each institution's existing Shariah artefacts per Section 8.4, and confirmation of the reporting formats the governance module must produce.

10. Glossary

Islamic commercial law

Term	Meaning as used in this document
Bay' al-dayn بيع الدين	Sale of debt. Selling a receivable at other than face value. Prohibited under the AAOIFI and wider Gulf position. The reason conventional supply chain finance cannot be adapted.
Bay' al-'inah بيع العينة	Sale followed by immediate buy-back to the original seller. Prohibited. Enforced against by counterparty distinctness checks.
Gharar غرر	Excessive uncertainty in a contract's subject matter, price or terms. Avoided by determinacy requirements before execution.
Hiyal حيل	Legal ruses — arrangements formally compliant but substantively equivalent to a prohibited transaction. Explicitly rejected as a design approach.
Ibra' إبراء	Rebate or waiver, as on early settlement of a fixed-price obligation.
Ijarah / Ijarah Muntahia Bittamleek إجارة	Lease; lease ending in transfer of ownership. The lessor bears ownership risk and major maintenance throughout.
Istisna'a استصناع	Contract for manufacture of a specified item, with staged payment. Used with a parallel contract for purchase order finance.
Kafala كفالة	Guarantee. A fee beyond actual cost is generally not permitted.
Maysir ميسر	Gambling or speculation. Avoided by excluding contingent pricing and optionality.
Mudaraba / Musharaka مضاربة / مشاركة	Profit-sharing partnership; joint partnership with shared profit and loss.
Murabaha مرابحة	Cost-plus sale in which the seller discloses the original cost and the markup. The core structure of Wasl Phase 1.
Qabd / Qabd hukmi قبض	Possession; constructive possession. A precondition of onward sale.
Qard Hasan قرض حسن	Benevolent loan repayable at face value with no return to the lender.
Rahn رهن	Pledge or collateral. The pledgee may not benefit from the pledged asset.
Riba ربا	Interest or usury. The prohibition from which most of this design follows.
Salam سلم	Forward purchase with full advance payment for future delivery of specified fungible goods.
Tawarruq تورق	Commodity-based cash generation. Organised tawarruq is contested and was ruled impermissible by the OIC International Islamic Fiqh Academy. Excluded from this product.
Ta'widh / Gharamah تعويض / غرامة	Compensation for actual proven damage; penalty. The penalty element is not institutional income.
Wa'd وعد	Unilateral promise, which may be binding on the promisor.
Wakala وكالة	Agency.
AAOIFI	Accounting and Auditing Organization for Islamic Financial Institutions. Its Shari'ah Standards are the principal reference for the structures in this document.

Term	Meaning as used in this document
SSB	Shariah Supervisory Board. The panel of scholars appointed by a financial institution to rule on the permissibility of its products. The governing authority for everything in this document. Each institution has its own ; a ruling binds only the institution whose Board issued it.
Divergence map	The Phase 0 artefact recording, question by question, each Board's position and the configuration key that absorbs any difference between them. See Section 3.11.4.

Saudi institutions, systems and regulation

Term	Meaning as used in this document
SAMA	Saudi Central Bank. Prudential and conduct regulator; issues the Cyber Security Framework and the Cloud Computing Regulatory Rules.
ZATCA / FATOORA	Zakat, Tax and Customs Authority, and its mandatory e-invoicing regime. The source of cleared, cryptographically stamped invoices that underpin both underwriting and Shariah evidence in this design.
Nafath	National single sign-on and identity assurance service.
Wathq	Ministry of Commerce business information service providing verified commercial registration data.
SIMAH / Bayan	Credit bureaux providing consumer and commercial credit information.
GOSI / Qiwa	Social insurance and labour platforms, used here for business-substance signals.
Monsha'at	Small and Medium Enterprises General Authority.
Kafalah	SME financing guarantee programme.
Etimad	Government procurement and contracting platform. Phase 3 integration for government contract finance.
SARIE	National payment system used for settlement.
PDPL / SDAIA	Personal Data Protection Law and the authority responsible for it.
NCDC	National Center for Digital Certification. Licenses the certification service providers whose certificates give a digital signature its statutory weight.
CR	Commercial Registration number. The verified legal entity identifier used for counterparty distinctness enforcement.

Technical

Term	Meaning as used in this document
PAdES	PDF Advanced Electronic Signatures. The signature profile used for executed instruments, with long-term validation so signatures remain verifiable after certificate expiry.
TSA / RFC 3161	Timestamping authority and its protocol. Provides the external clock against which leg sequence is proved.
HSM	Hardware security module. Custodian of the institutional sealing key.
Hash chain	Each executed document records its predecessor's hash, making the sequence verifiable from the documents alone, independently of the application's logs.

Term	Meaning as used in this document
Transactional outbox	Pattern ensuring an external side effect is never lost and never duplicated when a process fails between a database commit and an external call.
Gate	In this document, a compliance precondition enforced by the state machine. Not a validation rule; there is no path around it and no role that bypasses it.

IOTA Technologies

Sanad Platform · Wasl SME Supply Chain Finance

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This document is a systems design and carries no Shariah authority.

All structures and controls described are subject to the ruling of each deploying institution's Shariah Supervisory Board.